Total asphericity and exact abelian colimits

TOTEX2: proof audit and research record

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Abstract

This is an incomplete research record. We do not claim a proof or a counterexample for the implication from total asphericity to exact abelian colimits. We give complete proofs of the results explicitly labelled proved, distinguish the linear obstructions from their set-valued analogues, and record unresolved steps. A fully specified choice of repaired Adámek–Koubek–Reiterman targets gives a quasi-filtered category which is not total, deciding the strictness of that inclusion. No priority is claimed.

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A The main implication and coefficients

A.1 Scope and the precise unresolved question

We fix a universe and take all categories to be small in it. We use covariant functors $H : A \to \mathrm{Set}$, with arrows in $\mathrm{El}_A H$ directed by $(a, h) \xrightarrow{f} (b, H(f)h)$. We put $BA = |NA|$. A nonempty connected category A is total if each component of every such category of elements is weakly contractible. Exactness means exactness of $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$. The module $P_p = \mathbb{Z}[A(p, -)]$ is covariant; $\Delta\mathbb{Z}$ is the constant right $\mathbb{Z}A$-module. We write $T_i(M) = \mathrm{Tor}_i^{\mathbb{Z}A}(\Delta\mathbb{Z}, M) = L_i \mathrm{colim}_A M$.

Undecided statement A.1 (Open in this incomplete record: Part A.1)**.** We have not established whether every total category is exact. Equivalently, we have not proved that every finite parallel family $a_0, \dots, a_k : p \to q$ has an integral affine post-equalizer.

For $k = 1$ the assertion follows from the audited permutation-module calculation in Lemma B.2. For $k \geq 2$, the transposed presentation of Proposition B.5 detects the required simultaneous annihilator. None of the calculations of the row presentation identifies its kernel with this intersection. An affirmative answer for triples alone would still leave larger finite families: postcomposing by an affine equalizer creates affine arrows, to which the ordinary triple hypothesis does not directly apply.

A.2 An arithmetic reduction

Proposition A.2 (Proved: fieldwise detection of integral exactness)**.** *For every small category A, including the empty category and categories with several components, the following conditions are equivalent:*

1. $\mathrm{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact;*
2. $\mathrm{colim}_A : (\mathbb{Q}\text{-Mod})^A \to \mathbb{Q}\text{-Mod}$ *is exact and, for every prime p, the functor* $\mathrm{colim}_A : (\mathbb{F}_p\text{-Mod})^A \to \mathbb{F}_p\text{-Mod}$ *is exact.*

No totality hypothesis is needed. Consequently, Parts A.1 and A.2 taken together ask equivalent universal questions. A negative answer to A.1 necessarily gives a failure over $\mathbb{Q}$ or over at least one prime field, although the obstructing diagram need not remain the same after reduction.

Proof. The ordinary abelian-group colimit of a diagram of R-modules, for $R = \mathbb{Q}$ or $\mathbb{F}_p$, carries the unique compatible R-module structure, and its underlying abelian group is the ordinary abelian-group colimit. This follows from the direct-sum-and-relations construction, since the subgroup of relations is an R-submodule. Forgetting scalars is exact, so (1) implies (2).

Assume (2). We use the normalized bar complex

$$C_n(A; M) = \bigoplus_{a_0 \to \cdots \to a_n}^{\text{nondegenerate}} M(a_0),$$

with its usual face differential; its homology is $T_n(M)$. If M is a $\mathbb{Q}$- or $\mathbb{F}_p$-module diagram, this is, after restriction of scalars, exactly the bar complex computing derived colimits over that field. Thus such diagrams have $T_i(M) = 0$ for $i > 0$. This comparison is an equality of the specified complexes, not an assumption that arbitrary base change is flat.

We recall the three closure facts used here. A direct sum or filtered colimit of acyclic diagrams is acyclic because the bar complex commutes with these operations and both operations are exact in Ab. If $0 \to U \to V \to W \to 0$ is objectwise exact, the bar complexes

form a short exact sequence; their long exact homology sequence shows closure under extensions and under kernels of epimorphisms between acyclic diagrams.

Let $M : A \to \mathrm{Ab}$ be arbitrary. For each prime p, its p-primary torsion subfunctor is the union of $M[p^n]$, where $M[p^n](a) = \{m : p^n m = 0\}$. The quotient $M[p^n]/M[p^{n-1}]$ is an $\mathbb{F}_p$-module diagram. Induction on n, then filtered colimits, prove acyclicity of the p-primary subfunctor. The full torsion subfunctor M_{tor} is their direct sum and is therefore acyclic. Put $M_{\mathrm{tf}} = M/M_{\mathrm{tor}}$. It is objectwise torsion free, so the natural map

$$M_{\mathrm{tf}} \longrightarrow \mathbb{Q} \otimes_{\mathbb{Z}} M_{\mathrm{tf}}$$

is a monomorphism in Ab^A. Its cokernel is objectwise torsion: each element is a finite rational linear combination, and a common denominator kills its class. Both the middle diagram and this cokernel are acyclic. Closure under kernels of epimorphisms makes M_{tf} acyclic. Finally the sequence $0 \to M_{\mathrm{tor}} \to M \to M_{\mathrm{tf}} \to 0$ makes M acyclic. Therefore $T_1(M) = 0$ for every M, which is equivalent to exactness of the right-exact colimit functor. □

Undecided statement A.3 (Open in this incomplete record: Part A.2)**.** We have not established rational exactness for every total category, nor exactness over each $\mathbb{F}_p$. Proposition A.2 is a reduction, not a decision of either implication.

A.3 Proof and construction routes investigated

The accompanying report records the precise scope of each attempted route. The model-category observation that totality makes ordinary simplicial-set colimits homotopy invariant does not by itself identify the underlying set colimit with the abelian-group colimit. A free abelian cotriple resolution has permutation modules as its terms, but some face maps send a basis vector to a linear combination; it is not the linearization of a simplicial set diagram. We therefore do not infer acyclicity of its colimit complex from totality. The row/column formulas below isolate the unresolved first simultaneous-intersection instance.

Proposition A.4 (Refuted: universal membership of the column module in the proposed resolving class)**.** *For a small category A, let $\mathcal{R}_A$ be the smallest class of covariant integral modules containing every $\mathbb{Z}[H]$, for $H : A \to \mathrm{Set}$, and closed under extensions, kernels of epimorphisms between members, arbitrary direct sums, direct summands, filtered colimits, and $F \mapsto F \otimes_{\mathbb{Z}} \mathbb{Z}[H]$. Every member of $\mathcal{R}_A$ is objectwise torsion-free. There is a finite filtered category A and three parallel arrows $a_0, a_1, a_2 : p \to q$ for which*

$$Q_a = \mathrm{coker}(P_q \longrightarrow P_p^2,\ t \longmapsto (t(a_1 - a_0), t(a_2 - a_0))) \notin \mathcal{R}_A.$$

Thus the first proposed target of approach P1 cannot work uniformly, even for filtered categories. This does not refute P1's alternative of finding a different module with the same T_1, and it is not a separator between totality and exactness.

Proof. Permutation modules are objectwise free. Subgroups, extensions, direct sums and direct summands of torsion-free abelian groups are torsion-free. A filtered colimit is torsion-free because an equation $n[x] = 0$, $n \neq 0$, holds at some later stage, where it implies that the representative itself is zero. Tensoring with a free abelian group is a direct sum of copies. All operations are evaluated objectwise, so the class of objectwise torsion-free modules contains $\mathcal{R}_A$.

Here is the promised category completely. Its objects are p, q, x, t. Besides identities, its arrows are

$$a_0, a_1, a_2 : p \to q, \qquad g_0, g_1, g_2, g_3 : q \to x, \qquad c_0, c_1, c_2 : p \to x,$$

and one arrow $u_y : y \to t$ for each $y = p, q, x$. There are no other arrows. Identities compose as identities, every composition ending at t is the indicated unique arrow, and the other compositions are specified by

	a_0	a_1	a_2
g_0	c_0	c_1	c_2
g_1	c_0	c_2	c_1
g_2	c_1	c_0	c_2
g_3	c_1	c_2	c_0

Associativity holds: a composable triple of nonidentity arrows must follow $p \to q \to x \to t$, and both composites are u_p. All other associativity equations contain an identity. The object t is terminal, so the category is nonempty, connected and filtered. It is total directly: for any $H : A \to \mathrm{Set}$, the object (a, h) of its category of elements maps to $(t, H(u_a)h)$. Its connected component is exactly the fibre of this value in $H(t)$ and has that terminal object; the assertion is also valid if $H(t)$ is empty.

At x, $P_q(x) = \mathbb{Z}^4$ on the g_j and $P_p(x) = \mathbb{Z}^3$ on the c_i. The column map has its image in $\ker(\mathrm{aug} : \mathbb{Z}^3 \to \mathbb{Z})^2$, a rank-four direct summand of $\mathbb{Z}^6$. With bases $c_1 - c_0, c_2 - c_0$ in both augmentation kernels, its matrix is

$$D = \begin{pmatrix} 1 & 0 & -1 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & -1 \\ 1 & 0 & 1 & 0 \end{pmatrix}.$$

Its determinant is 3, and its upper-left 3×3 minor has determinant -1. Hence its Smith invariant factors are $1, 1, 1, 3$, and

$$Q_a(x) \cong \mathbb{Z}^2 \oplus \mathbb{Z}/3.$$

More explicitly, $v = (1, 0, 1, 0)^{\mathrm{t}}$ represents a nonzero element of order 3 in the reduced cokernel: one has $D(1, 1, -1, -1)^{\mathrm{t}} = 3v$, while the unique rational preimage of v is $(1, 1, -1, -1)^{\mathrm{t}}/3$, which is not integral. Thus Q_a cannot belong to $\mathcal{R}_A$. The program `code/algebra_column_torsion.py` reconstructs the category, checks all 52 composable pairs and 121 composable triples, verifies the identities and terminal object, and recomputes the Smith form and this order-three certificate. Its output is `code/algebra_column_torsion.out`. The original colimit is evaluation at t, so $T_1(Q_a) = 0$ in this example; no nonexactness is being inferred from its objectwise torsion. □

Proposition A.5 (Refuted: rational direct membership in P1)**.** *There is a small filtered category A and a triple $a_0, a_1, a_2 : p \to q$ such that its rational column module*

$$Q_a = \mathrm{coker}(\mathbb{Q}[A(q, -)] \longrightarrow \mathbb{Q}[A(p, -)]^2, \quad t \longmapsto (t(a_1 - a_0), t(a_2 - a_0)))$$

does not belong to the smallest class of rational diagrams containing all permutation modules and closed under extensions, kernels of epimorphisms, arbitrary direct sums, direct summands, filtered colimits, and tensoring with permutation modules. This strengthens the integral torsion obstruction: all values of this example are rational vector spaces. It does not refute the alternative in P1 of using another module with the same T_1. Moreover, the same category and triple give an integral column module with torsion-free values which lies outside the original integral resolving class.

Proof. First fix an automorphism $\sigma : v \to v$ in a category. For a rational diagram M, put $T_M = M(\sigma)$ and consider the condition

$$\ker(T_M^{2n} - 2^{2n} \mathrm{id}_{M(v)}) = 0 \qquad \text{for every integer } n \geq 1. \tag{1}$$

Every permutation module satisfies this condition. A finite-support eigenvector of a permutation has support a finite union of finite orbits. On an orbit of length r its eigenvalue must have rth power one; the eigenvalue 2^{2n} cannot occur.

Condition (1) is preserved by subobjects, extensions, arbitrary sums and summands. It is preserved by filtered colimits because evaluation at v and the displayed operator commute with them, and filtered colimits of rational vector spaces are exact. In particular it is preserved by kernels of epimorphisms.

We verify the additional tensor operation explicitly. Let P be the permutation of a set S and give $M(v) \otimes \mathbb{Q}[S]$ the automorphism $T_M \otimes P$. Fix $n \geq 1$ and a finite-support vector z in the kernel of $(T_M \otimes P)^{2n} - 2^{2n}$. Decompose its support into P^{2n}-orbits. On an infinite orbit, the eigenvector equation and invertibility of T_M propagate any nonzero coefficient indefinitely, contradicting finite support. On an orbit of length r, iterating the coefficient equation gives

$$(T_M^{2nr} - 2^{2nr})m = 0.$$

The old condition forces $m = 0$. Thus $z = 0$, which proves tensor closure. The use of all even powers is essential; injectivity of $T_M + 2$ alone would not suffice for this operation.

Now define A. Its objects are p, q, v, t, with t terminal. The only endomorphisms of p, q, t are identities, and $\mathrm{End}(v) = \{\sigma^n : n \in \mathbb{Z}\}$ is infinite cyclic. The other nonterminal hom-sets are

$$A(p,q) = \{a_0, a_1, a_2\}, \quad A(q,v) = \{h_{j,n} : j = 1, 2,\ n \in \mathbb{Z}\}, \quad A(p,v) = \{b_n : n \in \mathbb{Z}\}.$$

There are no further arrows except the unique arrows to t. Composition with σ^m adds m to the subscript, and

$$h_{j,n}a_i = b_{n+e_{j,i}}, \qquad (e_{1,0}, e_{1,1}, e_{1,2}) = (0,1,2), \quad (e_{2,0}, e_{2,1}, e_{2,2}) = (0,2,1).$$

All composites with terminal arrows are the unique terminal arrow. These formulas define an associative category: all nonidentity nonterminal associativity checks are associativity of integer addition or the equality $m + (n + e_{j,i}) = (m + n) + e_{j,i}$; terminal composites are unique. The object t makes A filtered.

Set $R = \mathbb{Q}[z, z^{-1}]$, using z to represent the action of σ. At v, the domain of the column presentation is R^2 and each of its two target factors is R. The presentation matrix is

$$D = \begin{pmatrix} z-1 & z^2-1 \\ z^2-1 & z-1 \end{pmatrix}.$$

The sum and difference bases diagonalize it over R, giving entries $(z-1)(z+2)$ and $-z(z-1)$. Since $-z$ is a unit,

$$Q_a(v) \cong R/((z-1)(z+2)) \oplus R/(z-1).$$

For an explicit nonzero -2 eigenvector, let ξ be the class of $(z-1, z-1)$. Then

$$(z+2)(z-1, z-1) = D(1,1),$$

so $(T_{Q_a} + 2)\xi = 0$ and consequently $(T_{Q_a}^2 - 4)\xi = 0$. The functional obtained by evaluating at $z = -2$ and summing the two coordinates annihilates both columns of D but takes value -6 on $(z-1, z-1)$. Thus $\xi \neq 0$.

The column module fails (1) for $n = 1$, whereas the entire specified resolving class satisfies it. Hence it does not belong to that class. Since t is terminal, $\mathrm{colim}_A Q_a = Q_a(t) = \mathbb{Q}^2$ and $T_i(Q_a) = 0$ for $i > 0$. This example concerns direct membership only, not the main implication.

Proved integral strengthening. The same invariant and closure argument work over $\mathbb{Z}$. For the tensor operation on an infinite permutation orbit, choose an extreme nonzero coefficient in its finite support. The coefficient equation one step beyond that support says that T_M^{2n} kills that coefficient, contrary to invertibility of T_M. This argument does not divide by 2; the finite-orbit argument is unchanged.

Let $R_{\mathbb{Z}} = \mathbb{Z}[z, z^{-1}]$. The integral column matrix at v is the same D, and

$$\det D = -z(z-1)^2(z+2) = -z^4 + 3z^2 - 2z.$$

It is nonzero over the domain $R_{\mathbb{Z}}$, so $0 \to R_{\mathbb{Z}}^2 \xrightarrow{D} R_{\mathbb{Z}}^2 \to Q_a(v) \to 0$ is exact. Its reduction modulo any prime p is also injective: the determinant has leading coefficient -1, so it remains nonzero in the domain $\mathbb{F}_p[z, z^{-1}]$. The two free abelian terms in this presentation therefore give

$$\operatorname{Tor}_1^{\mathbb{Z}}(\mathbb{Z}/p, Q_a(v)) = \ker(D \bmod p) = 0 \qquad \text{for every prime } p.$$

Equivalently, $Q_a(v)$ has no element killed by a prime and is torsion free. At p and t the column module is $\mathbb{Z}^2$; at q it is $\mathbb{Z}^5$, since the single presentation column has a coordinate equal to 1. Thus every value is torsion free. The same vector $\xi = [(z-1, z-1)]$ is killed by $z+2$ and is nonzero: evaluate at $z = -2$ in $\mathbb{Z}[1/2]$, then sum the coordinates, obtaining -6. It violates the integral invariant for $n = 1$. Hence the integral column module also fails membership in P1's full resolving class, although all its higher colimits vanish because t is terminal. □

Exact symbolic certificate. The program `finite_rational_p1.py` checks every composable triple of the finitely many arrow families using three independent integer exponent variables. Equality of the resulting affine expressions proves the displayed associativity identities for all integer exponents; no finite truncation of the category is substituted. It also verifies the rational change of basis, the two polynomial relations for ξ, and the nonzero detecting functional. The full tensor-closure argument is the orbit proof above, not a numerical test.

A.4 Ramsey classes and integral affine equalizers

The following argument is included with full proof. We do not claim priority for the implication; the literature search did not identify an earlier statement in these terms.

Theorem A.6 (Proved: a Ramsey criterion for integral exactness)**.** *Let A be a small category with finite hom-sets. Suppose that any two objects have a common successor. Suppose also that A has the following morphism Ramsey property: for any objects p, q and positive integer c there is an object r such that every coloring*

$$\chi : A(p, r) \longrightarrow \{1, \dots, c\}$$

admits $g : q \to r$ for which χ is constant on $g \circ A(p, q)$. Then A has exact colimits in abelian groups. More precisely, if $a_0, \dots, a_k : p \to q$ and $k \geq 1$, every target r with the stated Ramsey property for $p, q, c = (3k+1)^k$ admits an integral affine combination $t \in \mathbb{Z}[A(q, r)]$ satisfying $ta_0 = \cdots = ta_k$.

Proof. Fix the family and such a target. Put $X = A(p, r)$, $Y = A(q, r)$, and define the homomorphism between finite-rank free abelian groups

$$D : \mathbb{Z}[Y] \longrightarrow \mathbb{Z}[X]^k, \qquad D[g] = ([ga_i] - [ga_0])_{i=1}^k.$$

Let $\epsilon : \mathbb{Z}[Y] \to \mathbb{Z}$ be augmentation, and write $\epsilon(\ker D) = d\mathbb{Z}$ with $d \geq 0$. We must prove $d = 1$. Suppose otherwise. Choose

$$\theta = \begin{cases} 1/2, & d = 0, \\ \lfloor d/2 \rfloor / d, & d \geq 2. \end{cases}$$

Then $1/3 \leq \theta \leq 1/2$ and $d\theta = 0$ in the circle group $\mathbb{T} = \mathbb{R}/\mathbb{Z}$. Consequently

$$\phi : \operatorname{im} D \longrightarrow \mathbb{T}, \qquad \phi(Dt) = \theta\epsilon(t) \pmod{\mathbb{Z}}$$

is well-defined: two preimages differ by an element of $\ker D$, whose augmentation is a multiple of d.

This homomorphism extends to a homomorphism $\widetilde{\phi} : \mathbb{Z}[X]^k \to \mathbb{T}$. To see the extension directly, choose a Smith normal form for the subgroup $\operatorname{im} D$ of the finite-rank free group. In a suitable basis $e_1, \ldots, e_m$ it is generated by $s_1 e_1, \ldots, s_\ell e_\ell$, where $s_j > 0$. Choose an s_jth root in $\mathbb{T}$ of $\phi(s_j e_j)$ as the value on e_j, and assign arbitrary values to the remaining basis vectors. Such roots exist because multiplication by every positive integer on $\mathbb{T}$ is surjective.

Write $\lambda_i(x) \in [0, 1)$ for the representative of the value of $\widetilde{\phi}$ on the x-basis vector in block i. For every $g \in Y$, the construction gives

$$\sum_{i=1}^{k} (\lambda_i(ga_i) - \lambda_i(ga_0)) = \theta \pmod{\mathbb{Z}}. \tag{2}$$

Set $L = 3k + 1$ and color $x \in X$ by the k-tuple

$$(\lfloor L\lambda_1(x) \rfloor, \ldots, \lfloor L\lambda_k(x) \rfloor) \in \{0, \ldots, L-1\}^k.$$

Ramsey supplies $g : q \to r$ on whose p-subobjects this color is constant. In particular, ga_i and ga_0 have the same color for every i. Thus

$$\left| \sum_{i=1}^{k} (\lambda_i(ga_i) - \lambda_i(ga_0)) \right| < \frac{k}{3k+1} < \frac{1}{3}.$$

Every real representative of θ modulo $\mathbb{Z}$ has absolute value at least $1/3$, contradicting (2). Therefore $d = 1$, which means precisely that some $t \in \ker D$ has augmentation 1. A family of one arrow requires only 1_q. Together with the common-successor hypothesis, simultaneous integral affine equalization is Isbell–Mitchell's exactness criterion (Isbell and Mitchell, 1976, Theorem 3.1 and Proposition 3.3). □

Remark A.7 (Proved: scope of the finite-hom-set hypothesis)**.** It is used above only to give an elementary Smith normal form proof of extension to the circle. Divisibility makes $\mathbb{T}$ an injective abelian group, so the same argument works without finite hom-sets if one invokes that standard injectivity theorem. The finite version already covers the ordered-graph application below.

Corollary A.8 (Proved: ordered finite graphs are exact)**.** *Let $\mathcal{G}$ have one representative of each finite linearly ordered simple graph as objects and increasing induced graph embeddings as arrows, composed as functions. Then $\mathcal{G}$ has exact colimits in abelian groups. In particular it is not a counterexample to totality implying exactness.*

Proof. The category is small, has finite hom-sets, and has common successors: take the disjoint union of two graphs and concatenate their orders. Finite simple graphs with an arbitrary linear order have the Ramsey property by (Hubička and Nešetřil, 2019, Corollary 4.3, printed p. 51). The source defines embeddings by preservation and reflection of relations (printed p. 6), so these are induced embeddings in the graph case. Because a finite linear order is rigid, an isomorphic induced copy of p is specified by a unique embedding from p; hence its Ramsey property for copies is exactly the morphism Ramsey property used in Theorem A.6. That theorem applies. □

A.5 Explicit small ordered-graph certificates

Here q is the increasing induced path with vertex set $\{0 < 1 < 2\}$ and edge set $\{01, 12\}$. A triple (a, b, c) denotes the embedding of q with images $a < b < c$. These examples were first found by exact finite computation, then checked by the displayed integer identities. They are independent of the Ramsey theorem.

Proposition A.9 (Proved: three vertex maps)**.** *For the three embeddings v_0, v_1, v_2 of the one-vertex graph into q, an integral affine equalizer exists in the ordered graph r_6 on $\{0 < \cdots < 5\}$ with edge set*

$$\{01, 12, 13, 15, 23, 24, 34, 35\}.$$

It is

$$t = (012) - (015) + (124) - (134) + (235).$$

Proof. Each displayed triple has its consecutive pairs in the edge set and its outer pair absent, so it is an increasing induced embedding. The coefficient sum is $1 - 1 + 1 - 1 + 1 = 1$. The three vertex marginals are respectively

$$[0] - [0] + [1] - [1] + [2] = [2], \quad [1] - [1] + [2] - [3] + [3] = [2], \quad [2] - [5] + [4] - [4] + [5] = [2].$$

Thus $tv_0 = tv_1 = tv_2$ over $\mathbb{Z}$, and after change of coefficients over every unital coefficient ring. □

Proposition A.10 (Proved: two edge maps)**.** *For the embeddings e_0, e_1 of the ordered edge into the two edges of q, an integral affine equalizer exists in the ordered graph r_7 on $\{0 < \cdots < 6\}$ with edge set*

$$\{01, 12, 23, 35, 56, 14, 45\}.$$

It is

$$t = (012) + (123) + (235) + (356) - (014) - (145) - (456).$$

Proof. Each displayed triple is an increasing induced path by inspection of the specified edge set. The augmentation is $4 - 3 = 1$. In the free abelian group on embeddings of the edge, one has

$$(e_1 - e_0)_*((012) + (123) + (235) + (356)) = [56] - [01],$$
$$(e_1 - e_0)_*((014) + (145) + (456)) = [56] - [01].$$

Indeed the two telescoping sums are respectively

$$([12] - [01]) + ([23] - [12]) + ([35] - [23]) + ([56] - [35]),$$
$$([14] - [01]) + ([45] - [14]) + ([56] - [45]).$$

Subtracting gives $te_0 = te_1$. Precomposition with each of the two vertex maps into the edge also gives $tv_0 = tv_1 = tv_2$. □

Finite computations and their limits. The supplied script `finite_ordered_graphs.py` enumerates every graph on the fixed ordered vertex set $\{0, \ldots, n-1\}$ for $1 \le n \le 6$. It uses exact rational and finite-field arithmetic to compare the rank of D with the rank after adjoining augmentation as an extra row. The counts of targets admitting affine equalizers are zero for both families over all three fields when $n \le 5$. For $n = 6$, among all $2^{15} = 32768$ ordered graphs, the counts are

	$\mathbb{Q}$	$\mathbb{F}_2$	$\mathbb{F}_3$
three vertex maps	42	34	42
two edge maps	8	0	8

The first vertex certificate has ranks $5, 6$ over all three fields; the displayed seven-vertex edge certificate has ranks $6, 7$ over all three. A separate script, `finite_ordered_graphs_verify.py`, rebuilds the graph maps directly, checks all reported coefficients and marginals, and independently computes these ranks using `DomainMatrix` over the three exact coefficient domains. The exhaustive computation shows that the smallest target size for the vertex family over $\mathbb{Z}$ is 6, and for the edge family over $\mathbb{Z}$ is 7: an integral certificate reduces modulo 2, so the enumerated $\mathbb{F}_2$ obstructions give the latter lower bound. These minimal-size assertions are certified finite computations, not conceptual lower-bound proofs.

A separately recorded sample of 20000 seven-vertex graphs, using seed 20261113, is expressly not an exhaustive seven-vertex search. None of these finite searches by itself proves universal exactness or totality. The universal conclusion in Corollary A.8 instead follows from the circle-group argument and the opened structural Ramsey theorem.

A.6 A second homogeneous candidate: independent linear and cyclic orders

Proposition A.11 (Refuted: the linear–cyclic order candidate)**.** *Let C be a skeleton of the category of nonempty finite sets with an independent linear order and cyclic order, whose arrows are injections preserving both orders. Cyclic orders on one or two points are taken to be unique. This small category is neither total nor exact. More precisely, BC is simply connected and $H_2(BC;\mathbb{Z}) \cong \mathbb{Z}$, and there is a five-arrow parallel family with no affine equalizer over any nonzero commutative ring. No assertion that C is quasi-filtered is made.*

Proof. Take p to have two points and q to have linear order $1 < 2 < 3 < 4$ and cyclic order $(1, 3, 2, 4)$. Every increasing pair ij defines an arrow $a_{ij} : p \to q$. For any target r, choose a cut in its cyclic order, giving a linear reading of that circle. For an arrow $u : p \to r$, define $\chi(u) \in \{0, 1\}$ to be one if the larger of its two points in the given linear order of r occurs first in the circular reading, and zero otherwise. Extend χ linearly to $R[C(p, r)] \to R$ for any commutative ring R.

For every $g : q \to r$, the circular order induced on its image is $(g1, g3, g2, g4)$. Examining the four possible positions of the restricted cut gives

$$\chi(ga_{12}) + \chi(ga_{23}) - \chi(ga_{13}) = 1,$$
$$\chi(ga_{12}) + \chi(ga_{24}) - \chi(ga_{14}) = 0.$$

These identities can also be read from the negative orientation of the triple 123 and the positive orientation of 124. For $t \in R[C(q, r)]$ annihilating the differences of the five arrows

$$a_{12}, a_{23}, a_{13}, a_{24}, a_{14},$$

write c for the common value $\chi(ta_{ij})$. The first identity gives $c = \mathrm{aug}(t)$ and the second gives $c = 0$. Thus every simultaneous annihilator has augmentation zero, at every target. In particular there is no affine equalizer when $1_R \neq 0$. The column exact sequences of Proposition B.5, with R in place of $\mathbb{Z}$, give $T_1(Q_a) \cong R$ directly, without assuming condition I. In the integral case the smaller four-arrow family $a_{12}, a_{23}, a_{34}, a_{14}$ already has an obstruction:

$$\chi(ga_{12}) + \chi(ga_{23}) + \chi(ga_{34}) - \chi(ga_{14}) = 1$$

forces $2c = 1$ for an affine equalizer.

To check totality independently, let X_n be the set of cyclic orders on the linearly ordered set $\{0, \ldots, n\}$, with face maps given by deletion followed by increasing relabeling. This is a semisimplicial set. Its category of simplices is exactly C: an increasing injection is an arrow precisely when the cyclic order on its source is the restriction of that on its target. The usual simplex-category comparison gives $BC \simeq |X|$. One may obtain this comparison by realizing the

diagram of standard simplices over the simplex category. Its objectwise collapse to points is a homotopy-colimit equivalence, and its augmentation to $|X|$ is the simplex-category subdivision comparison. Equivalently, barycentric subdivision with the face occurrences retained gives the nerve of that category. This is the same semisimplicial comparison used in the finite sphere test, and does not identify the unrelated full cyclic category with C.

There is one zero-simplex, one one-simplex, two two-simplices, and six three-simplices. The cellular differentials in these degrees, with the alternating deletion convention, are

$$d_1 = 0, \qquad d_2 = (1,1) : \mathbb{Z}^2 \longrightarrow \mathbb{Z}, \qquad d_3 = 0 : \mathbb{Z}^6 \longrightarrow \mathbb{Z}^2.$$

For d_2, all three faces of either cyclic triangle are the unique edge, with coefficient sum $1 - 1 + 1 = 1$. For d_3, the six cyclic orders fixing the initial point have alternating triangle-face sums zero in each of the two orientation classes. The program `cyclic_linear_cuts.py` prints and checks all six columns, as well as all four cut identities above. It follows that $H_2(|X|;\mathbb{Z}) = \ker d_2 \cong \mathbb{Z}$. Its one-skeleton is a circle, and either two-simplex attaches with degree one on that circle. Hence $\pi_1(|X|) = 0$. Already the constant singleton C-set has element category C, whose nonzero second homology excludes totality. □

A.7 A compact inverse-limit reformulation of a stronger Ramsey condition

This subsection isolates a possible strengthening of totality. It does not establish that strengthening. A compact inverse-limit argument cannot be used to prove the main implication without supplying the missing hypothesis identified below.

Definition A.12 (Proved: agreement of the two Ramsey formulations for finite hom-sets)**.** A small category A has the finite-pattern morphism Ramsey property if, for every $p, q \in A$, finite subset $S \subset A(p,q)$, and positive integer c, there is $r \in A$ such that every coloring

$$\chi : A(p,r) \longrightarrow \{1, \dots, c\}$$

admits $g : q \to r$ for which χ is constant on gS. When all hom-sets are finite, this is equivalent to the morphism Ramsey property in Theorem A.6: take $S = A(p,q)$.

Theorem A.13 (Proved: compact inverse limits and finite-pattern Ramsey)**.** *Let A be a small nonempty category with common successors. The following are equivalent.*

1. *Every contravariant diagram $X : A^{\mathrm{op}} \to \mathrm{CompHaus}$ of nonempty compact Hausdorff spaces has a nonempty strict inverse limit.*
2. *A has the finite-pattern morphism Ramsey property.*

No bound on the hom-sets is assumed.

Proof. Suppose (2) fails for p, q, S, c. Necessarily S is nonempty. For each a, let $X(a)$ consist of the colorings of $A(p,a)$ that are nonconstant on gS for every $g : q \to a$. This is a closed subset of the compact product $\{1, \dots, c\}^{A(p,a)}$: each forbidden monochromaticity event concerns finitely many coordinates and is clopen. The failure of (2) says that every $X(a)$ is nonempty. Precomposition with $A(p,u)$, for $u : a \to b$, defines a continuous map $X(b) \to X(a)$, since a monochromatic gS after restriction would be a monochromatic $(ug)S$ before restriction. Thus X is a diagram as in (1). A compatible family $(\chi_a)_a$ would satisfy

$$\chi_a(h) = \chi_p(1_p) \qquad (h : p \to a).$$

In particular χ_q would be constant on S, contrary to its defining condition for $g = 1_q$. This proves (1) ⇒ (2).

For the converse, first observe that a Ramsey target has a finite witness set. If r works for p, q, S, c, the clopen sets

$$U_g = \{\chi : \chi \text{ is constant on } gS\}, \qquad g : q \to r,$$

cover the compact coloring space. A finite subcover gives a finite $G \subset A(q, r)$ such that some member of G witnesses every coloring.

It follows that finitely many finite-pattern requirements can be satisfied simultaneously. To spell this out, let $S_j \subset A(p_j, q)$ and c_j be given for $1 \le j \le m$. Put $r_0 = q$. At stage j apply (2) to the finite pattern

$$S_j^{(j-1)} = \bigcup_{h_i \in G_i,\, 1 \le i < j} h_{j-1} \cdots h_1 S_j \subset A(p_j, r_{j-1}),$$

where $S_1^{(0)} = S_1$. Obtain a target r_j and a finite witness set $G_j \subset A(r_{j-1}, r_j)$. Given colorings at r_m, choose $h_m \in G_m$ first, then $h_{m-1} \in G_{m-1}$ for the appropriate pullback coloring, and continue backwards. The composite $g = h_m \cdots h_1 : q \to r_m$ makes every gS_j monochromatic. Indeed the prefix $h_{j-1} \cdots h_1 S_j$ is contained in $S_j^{(j-1)}$, so the choice at stage j controls every possible subsequent choice of that prefix.

Now let X be as in (1). Its limit is cut out in the compact product $\prod_a X(a)$ by the closed equations

$$x_a = X(f)x_b \qquad (f : a \to b).$$

Fix finitely many such equations, indexed by a finite set E of arrows. For each source a in E, fix a neighborhood U_a of the diagonal in $X(a) \times X(a)$. Choose a finite cover of $X(a)$ by sets whose Cartesian squares lie in U_a, and assign each point to one containing cover member. Such a cover exists by the compatible uniformity of a compact Hausdorff space. This gives a finite coloring of $X(a)$; continuity of this coloring is not required.

Choose a common successor q of the finitely many objects involved and arrows $j_a : a \to q$. For each source a use the finite pattern

$$S_a = \{j_a\} \cup \{j_b f : (f : a \to b) \in E\}.$$

Apply the simultaneous Ramsey assertion with the cover color counts to obtain r. Choose $z \in X(r)$ and color $h : a \to r$ by the cover color of $X(h)z$. The resulting $g : q \to r$ makes the assignments $x_a = X(gj_a)z$ satisfy

$$(x_a, X(f)x_b) \in U_a \qquad ((f : a \to b) \in E).$$

Choose arbitrary values for the other coordinates. Thus every finite set of equations can be satisfied to arbitrary uniform accuracy. Use closed entourages for each equation: their inverse images in the product have the finite intersection property, while the intersection over all such entourages is exactly the equality condition. Compactness therefore supplies a point satisfying every equation. This proves (1). □

Corollary A.14 (Proved: the compact assertion implies integral exactness)**.** *A nonempty small category with common successors satisfying either condition of Theorem A.13 has exact abelian colimits.*

Proof. The proof of Theorem A.6 uses monochromaticity only on the specified finite family $\{a_0, \ldots, a_k\}$, so its Ramsey input can be replaced by Definition A.12. For arbitrary hom-sets, replace its finite-rank Smith normal form step by extension of a character into $\mathbb{R}/\mathbb{Z}$. For completeness, that extension needs only divisibility: choose a maximal partial extension on a subgroup by Zorn's lemma. If its domain is not the whole group, adjoin an element x. If a

least positive n has nx in the old domain, choose an nth root of its assigned value; if no such n exists, choose any value for x. This extends the character in either case, a contradiction. The circle argument then gives an integral affine equalizer for every finite parallel family.

To see directly that these equalizers suffice, two parallel affine arrows have finite supports with augmentation one. Equalizing all ordinary arrows in the union of those supports by a further affine arrow makes the two affine composites equal. Common successors and this property make $\mathrm{Aff}_{\mathbb{Z}}(A)$ filtered. The exactness criterion of Isbell and Mitchell (1976, Theorem 3.1 and Proposition 3.3) applies. □

Proposition A.15 (Proved: uniformly bounded finite fibers)**.** *Let A be nonempty, connected, and quasi-filtered. Every contravariant diagram $X : A^{\mathrm{op}} \to \mathrm{Set}$ of nonempty finite sets with $|X(a)| \le N$ for one finite integer N has a nonempty inverse limit.*

Proof. Let A_X have the objects of A, with parallel arrows identified precisely when they induce the same function $X(b) \to X(a)$. Functoriality makes this an equivalence relation compatible with composition. Conditions I and IV descend from A by choosing representatives, applying their required square or affine equality in A, and then passing to the quotient. Hence A_X is quasi-filtered. For each fixed a one has

$$|A_X(a,b)| \le |X(a)|^{|X(b)|} \le |X(a)|^N \qquad \text{for every } b.$$

By Isbell and Mitchell (1976, Proposition 4.5), every finite parallel family out of a is equalized by an ordinary arrow in A_X. Connectedness and I give common successors, so A_X is filtered.

The diagram X factors through A_X^{op}, with the same strict limit. Every finite set of compatibility equations for this diagram has a solution: filteredness gives a cocone on the finite diagram in question, and any point at its vertex supplies that solution. The compatibility conditions are closed in the compact product of the finite discrete values. They have the finite intersection property, so their intersection is nonempty. □

Example A.16 (Proved: coefficient pinning can destroy a colimit relation)**.** Let $A = \Delta_{\mathrm{inj}}$, which is exact and total, and let $B(n)$ be the free Boolean algebra on the generators $x_0, \dots, x_{n-1}$, also regarded as its associated Boolean ring with addition given by symmetric difference. An increasing injection $f : n \to m$ acts by $x_i \mapsto x_{f(i)}$. These are injective unital Boolean-algebra homomorphisms: the corresponding restriction maps on the sets of Boolean valuations are surjective.

In $B(2)$ the elements x_0, x_1 have the same class in $\mathrm{colim}_A^{\mathrm{Set}} UB$, since the two maps $1 \to 2$ carry the sole generator to them. Nevertheless, with $c = x_0$, the pairs (c, x_0) and (c, x_1) lie in different components of $\mathrm{El}_A(UB \times UB)$. Equality of the two coordinates is preserved and reflected by every structure map; it holds for the first pair and fails for the second.

The failure persists for the abelian colimit after pinning c. Consider the component E of $\mathrm{El}_A UB$ containing $(2, x_0)$. Its objects are exactly the pairs (n, x_j). Indeed renaming variables sends a Boolean expression to a generator only if that expression was itself a generator, as can be checked on valuations. On E, evaluation at the valuation which sends x_j to 1 and all other generators to 0 defines a natural homomorphism of additive groups

$$B(n) \longrightarrow \mathbb{F}_2.$$

For an arrow $f : (n, x_j) \to (m, x_{f(j)})$, the target valuation restricts to the specified source valuation, proving naturality. At $(2, x_0)$ this map takes x_0 to 1 and x_1 to 0. Consequently x_0 and x_1 remain distinct in the abelian colimit over E.

Thus moving to a category of elements in order to hold a multiplicative coefficient fixed need not preserve even a set-colimit relation available before that move. In a proposed Boolean-algebra argument, one must prove the needed lifted tuple relations; totality alone cannot be

cited for their existence. This example does not rule out a different homological argument exploiting the full Boolean identity.

Undecided statement A.17 (Open in this incomplete record: the missing compact implication)**.** We have not proved that totality implies either condition of Theorem A.13. Nor have we proved or refuted that exactness implies those conditions. Proposition A.15 only shows that a finite-valued obstruction for a quasi-filtered category must have unbounded fiber cardinalities. Signed measures obtained from affine equalizers cannot be replaced by genuine points without an additional argument.

A.8 Quasi-filteredness does not give compact convex inverse limits

The following result rules out a possible averaging shortcut in the AKR persistence problem. It concerns quasi-filteredness, including the nontotal choice in Theorem C.24; no totality assertion about other choices is used.

Proposition A.18 (Proved: a uniform obstruction to positive approximation)**.** *Let K^* be any repaired AKR category covered by Proposition C.4. For every object X and every finitely supported family of nonnegative real coefficients $(t_g)_{g:W\to X}$ with $\sum_g t_g = 1$, put*

$$\mu_i(v) = \sum_{g:\, gv_i = v} t_g \quad (v \in K^*(V,X),\ i = 1,2,3).$$

Then

$$\|\mu_2 - \mu_1\|_1 + \|\mu_3 - \mu_1\|_1 \geq 1.$$

In particular the initial triple has no arbitrarily accurate simultaneous equalizers obtained from probability combinations of ordinary arrows, even as the target varies.

Proof. We use the proved counting invariant from Proposition C.4: every nonempty finite set S of W-arrows into one object satisfies

$$\left|\bigcup_{g\in S}\{gv_1, gv_2, gv_3\}\right| > \tfrac{3}{2}|S|.$$

These point maps are distinct when the corresponding underlying points are distinct, since K^* is a concrete category of injections. For a point map v, set

$$w(v) = \max(\{t_g : gv_i = v \text{ for some } i\} \cup \{0\}).$$

Apply the counting invariant to the finite threshold sets $S_s = \{g : t_g > s\}$. Integrating the elementary identities for the lengths of the intervals $0 < s < t_g$ and $0 < s < w(v)$ gives

$$\sum_v w(v) = \int_0^\infty \left|\bigcup_{g\in S_s} g(W)\right| ds \geq \frac{3}{2}\int_0^\infty |S_s|\, ds = \frac{3}{2}.$$

All sums and integrals reduce to finite step functions on the support of t. Nonnegativity gives $w(v) \leq \max_i \mu_i(v)$. Moreover,

$$\max_i \mu_i(v) \leq \mu_1(v) + (\mu_2(v) - \mu_1(v))_+ + (\mu_3(v) - \mu_1(v))_+.$$

Every μ_i has total mass one. Thus the sums of the positive and negative parts of $\mu_i - \mu_1$ agree, and

$$\frac{3}{2} \leq \sum_v \max_i \mu_i(v) \leq 1 + \frac{1}{2}(\|\mu_2 - \mu_1\|_1 + \|\mu_3 - \mu_1\|_1).$$

Rearrangement proves the bound. No positive lower bound on $\inf_m a_m - 3/2$ is required. □

Theorem A.19 (Refuted: quasi-filteredness guarantees compact convex limits)**.** *There is a nonempty connected quasi-filtered small category A and a contravariant diagram of nonempty finite-dimensional compact convex sets, with continuous affine transition maps, whose strict inverse limit is empty. One may take $A = K^*$ as above.*

Proof. For an object X, write $E_X = K^*(W, X)$ and $P_X = K^*(V, X)$. Both sets are finite because the objects are finite sets and all arrows are injective functions. In the finite-dimensional real vector space $\mathbb{R}[P_X]^2$ put

$$D_X[g] = ([gv_2] - [gv_1],\, [gv_3] - [gv_1]) \qquad (g \in E_X).$$

If E_X is nonempty, Proposition A.18 says that the compact convex hull $C_X = \operatorname{conv}\{D_X[g] : g \in E_X\}$ does not meet the open unit ball for the ℓ^1 norm. Finite-dimensional convex separation gives a nonzero linear functional λ with

$$\inf_{z \in C_X} \lambda(z) \geq \sup_{\|b\|_1 < 1} \lambda(b) = \|\lambda\|_\infty.$$

Here one separates a convex set from a disjoint open convex set; the last equality is the elementary duality of the two specified norms. Dividing by $\|\lambda\|_\infty$ gives coefficient functions $\phi_1, \phi_2 : P_X \to [-1, 1]$ such that

$$\phi_1(gv_2) - \phi_1(gv_1) + \phi_2(gv_3) - \phi_2(gv_1) \geq 1 \qquad (g \in E_X). \tag{3}$$

Define $\mathcal{C}(X)$ to be the set of all pairs $(\phi_1, \phi_2) \in [-1, 1]^{2P_X}$ satisfying (3). It is a closed convex subset of a finite-dimensional cube and is nonempty by the preceding separation. If E_X is empty, the conditions are vacuous, so use the whole cube, which is nonempty also when P_X is empty.

For an arrow $u : X \to Y$, define

$$\mathcal{C}(u) : \mathcal{C}(Y) \longrightarrow \mathcal{C}(X), \qquad (\phi_i)_i \longmapsto \big(v \longmapsto \phi_i(uv)\big)_i.$$

This is continuous and affine. It preserves the inequalities because the inequality for $g : W \to X$ after restriction is the inequality for $ug : W \to Y$ before restriction. Restrictions compose in the required order, so $\mathcal{C}$ is the asserted contravariant diagram.

Suppose a compatible family $(\phi_{i,X})_{X,i}$ existed. For each $v : V \to X$, compatibility along v evaluated at 1_V would give

$$\phi_{i,X}(v) = \phi_{i,V}(1_V).$$

Hence each coefficient function is constant on all point maps, with the same constant across objects. At $X = W$, the inequality for $g = 1_W$ would then read $0 \geq 1$, a contradiction. Thus the limit is empty. Quasi-filteredness and connected nonemptiness of K^* are the established properties of the repaired AKR construction. □

Scope for the persistence problem. The theorem refutes the general compact convex inverse-limit lemma suggested as a way to extend locally bounded cycle functionals. A particular diagram of such functionals could still have a compatible point for additional reasons, but quasi-filteredness alone cannot provide it. The result neither decides whether K^* is total nor provides a total nonexact category. It also does not assume the stronger compact Hausdorff condition of Theorem A.13.

B Triples and the algebraic leads

Undecided statement B.1 (Open in this incomplete record: Part B.1)**.** We have not proved or refuted the assertion that every three parallel arrows in every total category admit an integral affine equalizer. The least undecided family size is three arrows. The equivalences below are exact obstruction formulas, not vanishing theorems for those obstructions. In particular, the vanishing of the row obstruction under totality does not establish the vanishing of the column obstruction.

B.1 Algebraic conventions and the audited input

All categories in this subsection are small. We write $P_x = \mathbb{Z}[A(x,-)] \in \mathrm{Ab}^A$ and $T_i(F) = L_i \operatorname{colim}_A F$. An ordinary arrow $f : x \to y$ induces the morphism of covariant modules $P_y \to P_x$, $t \mapsto tf$. The map $\epsilon_x : P_x \to \underline{\mathbb{Z}}$ is the natural augmentation. For $L \subseteq P_x$, the notation $\epsilon(L)$ means the image of $\operatorname{colim}_A L \to \operatorname{colim}_A P_x = \mathbb{Z}$; it does not assert that one object realizes every element of this image. Every P_x is projective and has colimit $\mathbb{Z}$, by the linear Yoneda lemma and the initial object $(x, 1_x)$ of $x\backslash A$. The comparison $T_i(\mathbb{Z}[H]) \cong H_i(B \operatorname{El}_A H; \mathbb{Z})$ is the isomorphism of the normalized bar and nerve chain complexes sending $(a_0 \to \cdots \to a_i, h)$ to the corresponding simplex of the category of elements. In particular, totality implies the acyclicity of these permutation modules, including when H is empty or has many components.

We read the proofs of TOTEX A.1–A.4 and B.1–B.4, rather than using their stated conclusions without their maps. The part of A.1 used here is precisely the preceding chain-complex identification. The part of A.4 used here is that a total category has common successors and satisfies I and IV. Its common-successor argument identifies the two values of the constant two-element action on the union of two disjoint principal cosieves; Mayer–Vietoris gives nonzero first homology of the resulting double. Applying this argument in the category of elements of $A(p,-)$ completes a span. Its pair argument applies the two short exact sequences below to the quotient by one family of binomial relations. These are the exact steps required here. We also opened Isbell and Mitchell (1976), Lemma 2.2, Proposition 2.3, Theorem 3.1 and Proposition 3.3. Lemma 2.2 reduces a finite zigzag in an indecomposable action by successive commuting squares. Proposition 2.3, for IV, then follows by equalizing the two arrows in the resulting cospan. Theorem 3.1 identifies exactness with componentwise filtered affinization, and Proposition 3.3 reduces its pair condition for affine arrows to a finite family of ordinary arrows. These are the versions, with covariant actions, used below.

Lemma B.2 (Proved: the row calculation used below)**.** *Let $a_0, \dots, a_k : p \to q$ be a finite family of ordinary arrows in an arbitrary small category, and let $z_i = a_i - a_0$ for $1 \le i \le k$. Put*

$$d : P_q^k \longrightarrow P_p, \qquad (t_i) \longmapsto \sum_i t_i z_i, \qquad K = \ker d,$$

and let H be $A(p,-)$ modulo all relations $ca_i \sim ca_0$. Then $\mathbb{Z}[H] = \operatorname{coker} d$ canonically, and

$$\begin{aligned} T_1(\mathbb{Z}[H]) &\cong \operatorname{coker}(\operatorname{colim} K \xrightarrow{\epsilon^k} \mathbb{Z}^k), \\ T_2(\mathbb{Z}[H]) &\cong \ker(\operatorname{colim} K \xrightarrow{\epsilon^k} \mathbb{Z}^k), \\ T_j(\mathbb{Z}[H]) &\cong T_{j-2}(K) \quad (j \ge 3). \end{aligned}$$

Consequently, for total A, ϵ^k is an isomorphism and K is colimit-acyclic. For $k = 1$ the module $J_i = P_q z_i \subseteq P_p$ is colimit-acyclic and has zero colimit.

Proof. At each object, the kernel of the map from the free group on a set to the free group on its quotient is generated by the differences defining the equivalence relation; a path of

elementary relations expresses each other difference as their sum. This identifies the displayed cokernel without any injectivity assumption on the action maps. Set $J = \operatorname{im} d$. Apply the long exact derived-colimit sequences to $0 \to K \to P_q^k \to J \to 0$ and $0 \to J \to P_p \to \mathbb{Z}[H] \to 0$. The map $\operatorname{colim} J \to \mathbb{Z}$ is zero, because its composite with the epimorphism $\mathbb{Z}^k \to \operatorname{colim} J$ is $\operatorname{colim} d = 0$. Exactness and the acyclicity of the representables give all three formulas. For total A the left sides vanish in positive degrees by the bar identification. In the one-row case, the same two exact sequences give $\operatorname{colim} J_i = 0$ and $T_j(J_i) = 0$ for $j > 0$. □

Proposition B.3 (Proved: Lead 1)**.** *Let A be total and $a_0, a_1, a_2 : p \to q$. Put $K_i = \ker(P_q \xrightarrow{z_i} P_p)$ and $L = K_1 \cap K_2$. Then the surjection in TOTEX B.1 is a canonical isomorphism*

$$T_1(P_q/(K_1 + K_2)) \;\cong\; \mathbb{Z}/\epsilon(L).$$

Proof. Lemma B.2 with one relation gives canonical augmentation isomorphisms $\operatorname{colim} K_i \cong \mathbb{Z}$. Let $S = K_1 + K_2 \subseteq P_q$. The exact sequence

$$0 \longrightarrow L \xrightarrow{\ell \mapsto (\ell, -\ell)} K_1 \oplus K_2 \xrightarrow{(u,v) \mapsto u+v} S \longrightarrow 0$$

identifies $\operatorname{colim} S$ with $\mathbb{Z}^2/\epsilon(L)(1,-1)$. The augmentation to $\operatorname{colim} P_q = \mathbb{Z}$ is induced by $(m,n) \mapsto m+n$. The sequence $0 \to S \to P_q \to P_q/S \to 0$ identifies its kernel with $T_1(P_q/S)$. Sending the class of $(m,-m)$ to $m \bmod \epsilon(L)$ gives the stated isomorphism. This is exactly the map defining the prior surjection, so the result proves its injectivity, rather than giving an unrelated abstract isomorphism.

For completeness, the second derivation in the prompt also type-checks, with its necessary sign made explicit. Put $J_i = P_q z_i \subseteq P_p$ and let $D = \{(tz_1, tz_2) : t \in P_q\} \subseteq J_1 \oplus J_2$. The map

$$P_q \longrightarrow (J_1 \oplus J_2)/D, \qquad t \longmapsto [(tz_1, 0)]$$

is onto: $[(uz_1, vz_2)] = [((u-v)z_1, 0)]$. Its kernel is $K_1 + K_2$, because $(tz_1, 0) = (wz_1, wz_2)$ means $w \in K_2$ and $t - w \in K_1$. Also $D \cong P_q/L$. Both J_i have zero colimit and are acyclic by Lemma B.2. The exact sequence $0 \to D \to J_1 \oplus J_2 \to (J_1 \oplus J_2)/D \to 0$ therefore gives $T_1(P_q/S) \cong \operatorname{colim} D = \mathbb{Z}/\epsilon(L)$. With the usual connecting-map convention, this second description using $t \mapsto [(tz_1, 0)]$ is the negative of the first augmentation description: a lift $u_1 + u_2 \in S$ maps to the diagonal element represented by u_2, whose augmentation is $-\epsilon(u_1)$. Using $t \mapsto [(-tz_1, 0)]$ makes the two specified isomorphisms agree. □

Proposition B.4 (Proved: Lead 2)**.** *Let A satisfy I; neither connectedness nor nonemptiness is required in this proposition. The following are equivalent:*

1. $\operatorname{colim}_A : \mathrm{Ab}^A \to \mathrm{Ab}$ *is exact;*
2. *for every $p, r \in A$ and every $z \in \mathbb{Z}[A(p,r)]$ with $\operatorname{aug} z = 0$, some affine $t : r \to s$ satisfies $tz = 0$.*

For every small A, every such p, r, z, and $L_z = \ker(P_r \xrightarrow{t \mapsto tz} P_p)$, one has canonically

$$T_1(P_p/P_r z) \cong \mathbb{Z}/\epsilon(L_z).$$

Proof. Set $J = P_r z$. The morphism $P_r \to P_p$ has zero map on colimits. The sequences $0 \to L_z \to P_r \to J \to 0$ and $0 \to J \to P_p \to P_p/J \to 0$ therefore give $T_1(P_p/J) \cong \operatorname{colim} J = \mathbb{Z}/\epsilon(L_z)$, just as in Lemma B.2. This formula does not use I.

If colimits are exact, the displayed quotient is zero. I makes the objects in each connected component have common successors: successively complete the spans along a finite connecting zigzag. All objects at which L_z is nonzero are in the component of r. A finite expression for

$1 \in \epsilon(L_z)$ can therefore be transported to one object and added there, producing an affine annihilator.

Conversely, if $\alpha, \beta : p \to r$ are affine arrows, their difference is such a z, so (2) equalizes every parallel affine pair. Each component of A has common successors by I, and hence so does its affinization. The pair condition makes every component of $\mathrm{Aff}_{\mathbb{Z}}(A)$ filtered. To verify the implication to exactness directly, extend an abelian diagram affinely. Its ordinary and affine colimits are canonically equal, since for $t = \sum n_i f_i$ of augmentation one, $x - tx = \sum n_i(x - f_i x)$ in the direct-sum presentation of the colimit. Each affine component has exact filtered colimits, and the colimit over all components is their direct sum, an exact functor in abelian groups. The empty category gives the zero functor and satisfies both conditions vacuously. □

Proposition B.5 (Proved: Lead 3)**.** *For every small category A, every integer $k \geq 0$ and every family $a_0, \ldots, a_k : p \to q$, put*

$$d^{\mathrm{t}} : P_q \longrightarrow P_p^k, \quad t \longmapsto (tz_i)_{i=1}^k, \qquad Q_a = \operatorname{coker} d^{\mathrm{t}}, \quad L = \ker d^{\mathrm{t}}.$$

Then canonically

$$\begin{aligned} T_1(Q_a) &\cong \mathbb{Z}/\epsilon(L), \\ T_2(Q_a) &\cong \ker(\operatorname{colim} L \to \mathbb{Z}), \\ T_i(Q_a) &\cong T_{i-2}(L) \qquad (i \geq 3). \end{aligned}$$

If objects in the component of q have common successors, the family has an affine equalizer if and only if $T_1(Q_a) = 0$. In the covariant-module presentation, Q_a has k generators x_i at p and one relation $\sum_i (a_i - a_0)x_i = 0$ at q.

Proof. Let $D = \operatorname{im} d^{\mathrm{t}}$. In the sequence $0 \to D \to P_p^k \to Q_a \to 0$, the map on colimits from D is zero: precompose it with the epimorphism $\mathbb{Z} = \operatorname{colim} P_q \to \operatorname{colim} D$, and obtain the zero column $\operatorname{colim} d^{\mathrm{t}}$. Thus $T_1(Q_a) = \operatorname{colim} D$ and $T_i(Q_a) = T_{i-1}(D)$ for $i \geq 2$. The exact sequence $0 \to L \to P_q \to D \to 0$ gives the three asserted formulas. The common-successor argument in the preceding proof converts augmentation image $\mathbb{Z}$ into one affine equalizer; the reverse implication is immediate. The presentation follows by sending the generator $1_q \in P_q(q)$ to the indicated vector in $P_p(q)^k$. For $k = 0$, $Q_a = 0$ and $L = P_q$, so every formula still holds and 1_q is the equalizer of the one-arrow family. The transposed row $P_q^k \to P_p$ has the permutation cokernel of Lemma B.2; no identification of these two cokernels is being used when $k > 1$. □

Proposition B.6 (Proved: the precise assertions of Lead 4)**.** *Suppose A is nonempty and connected and satisfies I and IV. For $M : A \to \mathrm{Ab}$ and $\alpha, \beta \in M(q)$, equality of their classes in $\operatorname{colim}_A^{\mathrm{Set}} UM$ implies that an affine $t : q \to s$ satisfies $t\alpha = t\beta$ in $M(s)$. Moreover A is exact if and only if, for every such M, q, α, β, equality in $\operatorname{colim}_A^{\mathrm{Ab}} M$ implies the same affine equalizability.*

Proof. The first equality puts the two elements in one connected component $X \subseteq UM$. We verify the part of Isbell and Mitchell (1976), Lemma 2.2 and Proposition 2.3, being used. Condition I reduces a zigzag in X to a cospan: at each reduction, a span of ordinary arrows is replaced by a commuting square. Thus some $b, b' : q \to r$ satisfy $b\alpha = b'\beta$ in $X(r)$. Condition IV gives an affine $c : r \to s$ with $cb = cb'$ in $\mathbb{Z}[A(q, s)]$. Put $t = cb = cb'$. Linearity of M yields $t\alpha = c(b\alpha) = c(b'\beta) = t\beta$.

If A is exact, Theorem 3.1 of the opened paper makes its affinization filtered. The equality of ordinary and affine abelian colimits was proved in Proposition B.4. In a filtered colimit of groups, two elements at the same object become equal under one transition, which here is an affine arrow. Conversely apply the asserted property to $M = P_p$ and to any two affine arrows $\alpha, \beta : p \to q$. Their classes in $\operatorname{colim} P_p = \mathbb{Z}$ are both 1, so the affine category equalizes every parallel pair. Common successors and Proposition B.4's argument prove exactness. □

Remark B.7 (Proved: scope of the final sentence of Lead 4)**.** The comparison map

$$\operatorname{colim}_A^{\mathrm{Set}} UM \longrightarrow U \operatorname{colim}_A^{\mathrm{Ab}} M$$

is surjective under the common-successor hypothesis: transport the finite sum representing an abelian colimit element to one object and add there. The two assertions above precisely locate the remaining issue in its fibres: set equality is already sufficient, while abelian equality may identify additional components. This does not identify exactness with finite-product preservation of the set-colimit functor. For example, for $A = \Delta_{\mathrm{inj}}$, $p = [0]$, and $M = P_p$, the elements $2[a] - [b]$ for distinct $a, b : p \to q$ and $[a]$ have the same abelian-colimit class. Ordinary increasing injections preserve the number and coefficients of nonzero basis terms, so these two elements lie in different set-colimit components. Nevertheless A is exact, by the already audited Isbell example. Thus the final phrase of Lead 4 is valid as motivation for the difference between the two colimits, and would be false if read as the asserted equivalence with siftedness or with injectivity of this comparison.

Lemma B.8 (Proved: a strengthened normal form for zero classes)**.** *Let A be a nonempty category in which every pair of objects has a common successor, and let $F : A \to \mathrm{Ab}$. For $\kappa \in F(s)$, its colimit class is zero if and only if there are an ordinary arrow $u : s \to y$, finitely many ordinary parallel pairs $e_j, e'_j : x_j \rightrightarrows y$, and $k_j \in F(x_j)$, such that*

$$F(u)\kappa = \sum_j (F(e_j) - F(e'_j))k_j \quad \text{in } F(y).$$

In particular Lead 5(ii) holds; its condition I can be weakened to common successors.

Proof. Define the subgroup

$$N_y = \sum_{x \in A} \sum_{e,e' \in A(x,y)} \operatorname{im}(F(e) - F(e')) \subseteq F(y).$$

Postcomposition sends N_y into N_z, so N is a subfunctor. Any parallel arrows $y \rightrightarrows z$ induce the same map $F(y)/N_y \to F(z)/N_z$, by the definition of N_z. Consequently F/N factors through the reachability preorder D: its objects are those of A, and it has exactly one arrow $y \to z$ when $A(y, z) \neq \varnothing$. Common successors make D filtered. Every element of every N_y dies in $\operatorname{colim}_A F$, and the universal properties therefore identify

$$\operatorname{colim}_A F \cong \operatorname{colim}_A (F/N) \cong \operatorname{colim}_D \overline{F}$$

canonically. If κ dies, its image in $\overline{F}(s)$ becomes zero at one object y of the filtered diagram $\overline{F}$. Choose any arrow $u : s \to y$ in A. Then $F(u)\kappa \in N_y$, and membership in a sum of subgroups supplies the finite expression claimed; coefficients can be absorbed into k_j. Conversely every summand on the right has zero colimit class, and an arrow preserves that class. The empty sum is allowed. □

Proposition B.9 (Proved: Lead 5(i), (iii), and (iv))**.** *Let A be total and let $a_0, a_1, a_2 : p \to q$. Use z_i, K_i, K, L as in Lemma B.2 and Proposition B.3, and put $K_3 = \ker(P_q \xrightarrow{a_2 - a_1} P_p)$. Choose affine pair equalizers $\rho_1 \in K_1(s)$, $\rho_2 \in K_2(s)$ and $\rho_3 \in K_3(s)$ at a common object s. Then*

$$\kappa = (\rho_1 - \rho_3, \rho_3 - \rho_2) \in K(s)$$

has zero class in $\operatorname{colim} K$. If an affine $t : s \to r$ kills κ, then $t\rho_1 = t\rho_2 = t\rho_3$ is an affine equalizer of the triple. Such a t exists whenever the finitely many pairs in a normal form of

Lemma B.8 have a common affine equalizer. In particular, a normal form with at most one summand implies the existence of such a t. Finally the natural morphism

$$\psi : K_1 \oplus K_2 \oplus K_3 \longrightarrow K, \qquad (k_1, k_2, k_3) \longmapsto (k_1 - k_3, k_2 + k_3)$$

has kernel canonically isomorphic to L, and

$$T_1(K/\operatorname{im}\psi) \cong \mathbb{Z}/\epsilon(L).$$

Proof. The row applied to κ is $\rho_1 z_1 - \rho_2 z_2 + \rho_3(z_2 - z_1) = 0$. Both coordinates of κ have augmentation zero, and Lemma B.2 identifies $\operatorname{colim} K$ with $\mathbb{Z}^2$ by those augmentations. Hence its colimit class vanishes, proving (i). If $t\kappa = 0$, equality of its first and second coordinates gives $t\rho_1 = t\rho_3 = t\rho_2$; the common affine arrow belongs to $K_1(r) \cap K_2(r)$, proving the first assertion in (iii). For a normal form $u\kappa = \sum(e_j - e'_j)k_j$, a common affine equalizer v of the pairs gives $(vu)\kappa = 0$. Condition IV gives such a v for one pair. No assertion is made that IV alone equalizes a finite list of pairs.

The displayed formula for ψ has image in K, because $-k_3 z_1 + k_3 z_2 = k_3(a_2 - a_1) = 0$. Its kernel consists precisely of $(\ell, -\ell, \ell)$ with $\ell \in L$; membership in K_3 follows from membership in $K_1 \cap K_2$. Write $S = K_1 \oplus K_2 \oplus K_3$, $I = \operatorname{im}\psi$ and $C = K/I$. Each K_i is colimit-acyclic with augmentation colimit $\mathbb{Z}$, by the one-relation case of Lemma B.2, and K is colimit-acyclic with colimit $\mathbb{Z}^2$. Right exactness applied to $0 \to L \to S \to I \to 0$ gives

$$\operatorname{colim} I = \mathbb{Z}^3/\epsilon(L)(1, -1, 1).$$

The map to $\operatorname{colim} K = \mathbb{Z}^2$ is induced by $(a, b, c) \mapsto (a - c, b + c)$. Its kernel is $\mathbb{Z}(1, -1, 1)/\epsilon(L)(1, -1, 1)$. The exact sequence $0 \to I \to K \to C \to 0$ identifies this kernel with $T_1(C)$, because $T_1(K) = 0$.

For the graph interpretation, at each x take a bipartite multigraph with vertex set $A(q, x) \amalg A(p, x)$ and, for each $g : q \to x$, three labelled edges $E_{g,i} : g \to ga_i$, $i = 0, 1, 2$. Distinct edges remain distinct even when their endpoints coincide. The homomorphism from $K(x)$ to the integral graph cycle group is

$$(t_1, t_2) \longmapsto \sum_g \left(-(t_{1,g} + t_{2,g})E_{g,0} + t_{1,g}E_{g,1} + t_{2,g}E_{g,2}\right).$$

The boundary at a vertex g is zero, and the boundaries at the other vertices are exactly $t_1 z_1 + t_2 z_2$. Conversely the boundary equation at each g expresses its edge-zero coefficient in terms of the other two, proving this is an isomorphism. It is natural under postcomposition, including maps that identify vertices or edges. The three two-colour cycle groups give respectively $(k_1, 0)$, $(0, k_2)$ and $(-k_3, k_3)$, whose sum is exactly $\operatorname{im}\psi$. Thus the graph description is literal over $\mathbb{Z}$, not just an analogy over a field. □

Corollary B.10 (Proved: what the mixed-cycle obstruction says)**.** *Under the hypotheses of Proposition B.9, suppose the triple has no affine equalizer. Its class $[\kappa]$ in $\operatorname{colim} I$ is the nonzero generator of $\ker(\operatorname{colim} I \to \operatorname{colim} K) \cong \mathbb{Z}/\epsilon(L)$. It cannot have a normal form in which every k_j belongs to $I(x_j)$, and it cannot have a normal form whose pairs have a common affine equalizer. In particular every one-term normal form is impossible.*

Proof. The element κ is $\psi(\rho_1, -\rho_2, \rho_3)$; its three augmentations are $(1, -1, 1)$. This gives the asserted generator. The absence of an affine equalizer means $\epsilon(L) \neq \mathbb{Z}$, using common successors, so that generator is nonzero. A normal form with coefficients in I would kill its class already in $\operatorname{colim} I$, a contradiction. The second assertion follows from Proposition B.9(iii). Thus "carried around a 1-cycle" must mean a bar chain with coefficients in K, or equivalently the connecting class in $T_1(K/I)$, rather than a nontrivial ordinary homology class of BA: the latter is contractible for a total category. □

C The category construction and finite tests

Undecided statement C.1 (Open in this incomplete record: Part C.1)**.** For the repaired AKR construction and every admissible choice of its parameters, the requested assertion is

$$H_2\big(\mathrm{El}_{K^*}\big(K^*(V,-)/(cv_1 \sim cv_2)\big);\mathbb{Z}\big) \neq 0.$$

We have not established this universally quantified assertion or its negation. Theorem C.24 proves infinite-stage survival for a fully specified choice of sufficiently large targets, and more generally for such choices over any admissible parameter sequence. It does not cover every admissible target choice. The computations concern explicitly identified finite substages and do not extend that quantifier. The all-coefficient obstruction in Proposition C.4 establishes nonexactness independently of this unresolved totality question.

C.1 The link comparison in Lead 7

Proposition C.2 (Proved: the link comparison, with its necessary hypothesis)**.** *Let A and D be small categories, let $u : D \to A$ be a functor, and let*

$$R = \operatorname*{colim}_{d\in D} A(-,u(d)) : A^{\mathrm{op}} \longrightarrow \mathrm{Set}.$$

Assume that every connected component of every comma category $x \downarrow u$ is weakly contractible. Form A' by adjoining one object Z, by setting $A'(x,Z) = R(x)$ for $x \in A$, by allowing no arrows from Z to A, and by setting $A'(Z,Z) = \{1_Z\}$; composition with arrows of A is the presheaf action of R. Let $H' : A' \to \mathrm{Set}$ be any covariant functor, and put $H = H'|_A$.

Define the two-sided category $E(R,H)$ to have objects (x,r,k) with $x \in A$, $r \in R(x)$ and $k \in H(x)$. An arrow $(x,r,k) \to (y,s,l)$ is an arrow $b : x \to y$ of A satisfying $R(b)s = r$ and $H(b)k = l$. Then the functor

$$F : \mathrm{El}_D(u^*H) \longrightarrow E(R,H), \qquad (d,k) \longmapsto (u(d),[d,1_{u(d)}],k)$$

which sends an arrow $\delta : d \to d'$ to $u(\delta)$ is a weak equivalence on nerves. For every $h \in H'(Z)$ the old comma link

$$L_h = (\mathrm{El}_A H) \downarrow (Z,h)$$

is a full union of connected components of $E(R,H)$. If $e = \dim ND < \infty$, then $H_j(BL_h;\mathbb{Z}) = 0$ for $j > e$ and the inclusion induces an injection

$$H_j(B\,\mathrm{El}_A H;\mathbb{Z}) \longrightarrow H_j(B\,\mathrm{El}_{A'} H';\mathbb{Z}) \quad (j > e).$$

The assertion also holds with arbitrary constant coefficients. If D is empty, R and all links are empty and adjoining Z adds only isolated objects to the category of elements.

Proof. For any $x \in A$, the set $R(x)$ is canonically $\pi_0(x \downarrow u)$: its representatives are pairs $(d, b : x \to u(d))$, and its generating identifications are $(d,b) \sim (d', u(\delta)b)$ for $\delta : d \to d'$. Fix $w = (x,r,k) \in E(R,H)$. An object of $w \downarrow F$ is specified by $d \in D$, $l \in H(u(d))$ and $b : x \to u(d)$ with $[d,b] = r$ and $H(b)k = l$. Thus l is forced. Sending this object to (d,b) gives an isomorphism of categories

$$w \downarrow F \;\cong\; (x \downarrow u)_r,$$

where the right side is the connected component labelled by r. Indeed, an arrow on either side is exactly a $\delta : d \to d'$ with $u(\delta)b = b'$. The assumed weak contractibility proves the nerve comparison as follows. For a general functor $F : X \to Y$ with these comma nerves weakly contractible, form the bisimplicial set

$$B_{p,q} = \{(y_0 \to \cdots \to y_p \to F(x_0),\ x_0 \to \cdots \to x_q)\}.$$

For a fixed q-simplex of NX, its horizontal fiber is $N(Y \downarrow F(x_0))$, a nerve with terminal object $(F(x_0), 1)$. For a fixed p-simplex of NY, its vertical fiber is $N(y_p \downarrow F)$, weakly contractible by hypothesis. Realizing first in either direction therefore shows that the projections from the diagonal realization to BX and BY are weak equivalences. Here the levelwise statement passes to realization because degeneracy inclusions of simplicial sets are cellular inclusions; the usual skeletal filtration proves the assertion by successive gluing along those inclusions. The two maps to BY, one by its projection and one by BF after projection to BX, are homotopic: the joining arrow $y_p \to F(x_0)$ is the joining edge of the displayed chain and supplies the simplicial mapping-cylinder homotopy. Thus BF is a weak equivalence. Applying this argument to the comma categories just computed proves the claimed comparison.

The function

$$\lambda : \operatorname{Ob} E(R, H) \longrightarrow H'(Z), \qquad (x, r, k) \longmapsto H'(r)k$$

is constant on every arrow, by functoriality. The full subcategory $\lambda^{-1}(h)$ is exactly L_h, including its arrows. It is consequently a union of connected components. The restriction of F to the union of source components mapping into L_h remains a weak equivalence: the same comma-category computation applies separately on these components. The projection $\mathrm{El}_D(u^*H) \to D$ is a discrete opfibration. A nonidentity arrow cannot project to an identity, so a nondegenerate simplex in its nerve has dimension at most e. This gives the homology vanishing for every link.

For completeness, the attachment assertion is a chain-level statement, not an assumption that a link embeds as a subcategory of the old category. Every nondegenerate simplex of $N\,\mathrm{El}_{A'} H'$ that is not old ends at exactly one (Z, h); deleting this final vertex gives a simplex of the corresponding comma link, including the final arrow. Hence the normalized relative chain complex is canonically

$$C_*(N\,\mathrm{El}_{A'} H', N\,\mathrm{El}_A H) \cong \bigoplus_{h \in H'(Z)} \widetilde{C}_{*-1}(NL_h).$$

Its long exact sequence contains

$$\bigoplus_h H_j(BL_h) \longrightarrow H_j(B\,\mathrm{El}_A H) \longrightarrow H_j(B\,\mathrm{El}_{A'} H'),$$

and the first group is zero for $j > e$. All chains have finite support, so the direct sum and the argument remain valid for infinite sets of new elements and for arbitrary small indexing categories. □

Proposition C.3 (Proved: the equalizing-precomposition obstruction)**.** *Consider a cyclic quasi-filter indexing graph with copies of S at its vertices and copies of T on its edges, the two maps on each edge being chosen from a parallel pair $f, g : T \rightrightarrows S$. If an arrow $c : x \to T$ satisfies $fc = gc$, the component of $x \downarrow u$ containing these lifts has nonzero first integral homology. If f and g are monomorphisms, this component is precisely a subdivided circle. Consequently the comma hypothesis in Proposition C.2 fails. In the repaired first-stage AKR quasi-filter for v_1, v_2, the last and first maps satisfy $\alpha_{4k} v_1 = \alpha_0 v_1$, so this obstruction occurs when quasi-filtering (α_0, α_{4k}) at the next stage.*

Proof. At every vertex choose the arrow $fc = gc : x \to S$, and at every edge choose $c : x \to T$. These choices form a copy of the subdivided indexing circle in $x \downarrow u$. This comma category has no nonidentity composable pair, because its projection to the bipartite indexing graph has none. Its nerve is therefore a graph; the displayed cycle cannot be the boundary of a two-chain and has nonzero first homology. If f and g are monomorphisms, either map $A(x, T) \to A(x, S)$ is injective. The vertices of the displayed cycle then have no additional incident edges, and

its component is exactly that circle. The final application is the last neighboring equality in the repaired formula, with $c = v_1$. In a category where f, g are not monomorphisms the component can have further branches or cycles; only the noncontractibility, not literal equality with a circle, is automatic. □

What this does not prove about the infinite AKR category. The dimension-one conclusion applies only when the strict colimit of representables has the stated comma-category comparison. The preceding proposition supplies explicit AKR attachments where that hypothesis fails. Thus it cannot, by itself, establish persistence of the first-stage two-dimensional class. In particular we do not infer $H_2(L_h) = 0$ for those exceptional links from the dimension of the quasi-filter indexing graph.

Proposition C.4 (Proved: all-ring consequence of the known counting invariant)**.** *For every admissible choice in the repaired AKR Construction C.1, for every nonzero unital commutative ring R, and for every object X of K^*, the R-linear map*

$$R[K^*(W, X)] \longrightarrow R[K^*(V, X)]^2, \qquad t \longmapsto \big(t(v_2 - v_1), t(v_3 - v_1)\big)$$

is injective. In particular, the three initial point maps have no R-affine equalizer. The same injectivity holds for finitely supported coefficients in an arbitrary abelian group, with the evident meaning of the two differences.

Proof. We use the counting invariant of the opened AKR Theorem C.2, pp. 385–389, with the repaired maps satisfying its required (9)–(11). We checked the induction, not merely its conclusion. Its statements are: for an object X created by stage m and an arrow $g : Y \to X$,

1. if a W-arrow $h : W \to X$ has at least two image points in the image of g, it factors through g;
2. if $h_1, \dots, h_r : W \to X$ are distinct and each meets the image of g in at most one point, then their union outside that image has at least $a_m r$ points.

Here $3/2 < a_m \leq 2$. To check the source's inductive use of these statements, write every incoming map at a new quasi-filter target as α_i followed by an old map. Property (10) moves any overlap of two copies through consecutive copies, and (11) identifies each consecutive overlap with the image of one of the old pair. The first inductive statement then routes the entire W-arrow through the neighbor, proving the new first statement. For the second statement, partition the W-arrows by their first available copy, discard a block of $3n$ consecutive indices of least cardinality, and sweep the remaining indices from the two ends toward that block. Every newly counted W-arrow meets the preceding overlap in at most one point: otherwise the just-proved routing statement would have assigned it to an earlier copy. The old second statement thus adds at least a_m points per retained arrow. The discarded fraction is at most $3n/s \leq 1 - a_{m+1}/a_m$, giving $a_{m+1}r$. A disjoint-sum target separates the two sets of arrows; the same inequalities apply separately to its summands. These are precisely the induction steps on pp. 386–389, and all use of overlaps is justified by the repaired (10)–(11), not the invalid restriction assertion.

For clarity, the source's displayed set $L = \{1, \dots, n\} \setminus \widetilde{H}_{j_0}$ on p. 387 must read $\{1, \dots, r\}$; all surrounding cardinality formulas use the latter set. This is an index typo, not an additional hypothesis.

Fix now a nonempty finite set S of r distinct W-arrows into X. Two distinct such arrows meet in at most one point: otherwise the first statement would factor one through the other

via an endomorphism of W, and $K^*(W,W) = \{1_W\}$. Applying the second statement with one member of S as g therefore gives

$$\left|\bigcup_{h\in S} h(W)\right| \geq a_m(r-1)+3 > \frac{3}{2}r.$$

This is also valid for $r = 1$. There are exactly $3r$ incidences (h, i) with $h \in S$ and $i \in \{1, 2, 3\}$. If each image point had at least two incidences, the displayed union would have at most $3r/2$ points. Hence some image point has exactly one incidence.

Let $t = \sum_{h\in S} c_h h$ have all $c_h \neq 0$, and choose that private point, say $h_0 v_i$. Its coefficient in tv_i is c_{h_0}, whereas its coefficient in tv_j is zero for either $j \neq i$. Thus the three vectors tv_1, tv_2, tv_3 cannot be equal. This uses no division and no restriction on the characteristic or on zero divisors. It proves the claimed injectivity for coefficients in any abelian group and hence for every specified ring. In particular, it applies to $\mathbb{Q}$ and every $\mathbb{F}_p$ simultaneously. □

Proposition C.5 (Proved: an exact cycle-space test for one attachment)**.** *Let A' be obtained from A by adjoining one new maximal object Z with identity endomorphism only. Suppose a family $\alpha_i : T_i \to Z$ ($i \in I$, possibly infinite) generates all incoming arrows: every $b : X \to Z$, $X \in A$, can be written $\alpha_i c$ with $c : X \to T_i$ in A. For any covariant functor $F' : A' \to \mathrm{Ab}$, write $F = F'|_A$ and $Q = \mathrm{colim}_A F$. Define homomorphisms*

$$a : \bigoplus_{i\in I} F(T_i) \longrightarrow F'(Z), \quad (x_i) \longmapsto \sum_i F'(\alpha_i)x_i, \qquad b : \bigoplus_{i\in I} F(T_i) \longrightarrow Q, \quad (x_i) \longmapsto \sum_i [x_i].$$

Then, canonically,

$$\mathrm{colim}_{A'} F' \cong (Q \oplus F'(Z))/\{(b(x), -a(x)) : x \in \bigoplus_i F(T_i)\},$$

$$\ker(Q \to \mathrm{colim}_{A'} F') = b(\ker a).$$

For the AKR action $H = h_V/(cv_1 \sim cv_2)$, put $K(X) = \ker(\mathbb{Z}[A(W,X)] \xrightarrow{,v_2-v_1,} \mathbb{Z}[A(V,X)])$. Thus $K(X)$ is the integral cycle group of the graph whose vertices are V-arrows and whose edges are W-arrows, with edge c directed from cv_1 to cv_2. For a quasi-filter attachment all $T_i = T$. Its old-to-new map on $H_2(B\,\mathrm{El}\,H)$ is injective if and only if

$$\sum_i \alpha_i \kappa_i = 0 \text{ in } K(Z) \quad \Longrightarrow \quad \left[\sum_i \kappa_i\right] = 0 \text{ in } \mathrm{colim}_A K$$

for every finitely supported family $\kappa_i \in K(T)$.

Proof. A cocone from F' to an abelian group M is an old cocone $Q \to M$ together with a map $F'(Z) \to M$ agreeing along all incoming arrows. Agreement along α_i suffices: for $b = \alpha_i c$, naturality of the old cocone gives agreement along c first, hence along b. This proves the displayed pushout presentation. The class of $(q, 0)$ is zero precisely when $(q, 0) = (b(x), -a(x))$ for some x, which proves the kernel formula. No ordinary single-object representative for a general colimit element is assumed here; the direct sum explicitly allows finite sums at several objects.

For the last assertion, linearizing the binary equivalence relation gives

$$0 \longrightarrow K \longrightarrow P_W \xrightarrow{,v_2-v_1,} P_V \longrightarrow \mathbb{Z}[H] \longrightarrow 0.$$

Split this at the image of the middle map. The projectivity of P_W and P_V, together with $\mathrm{colim}\, P_W = \mathrm{colim}\, P_V = \mathbb{Z}$ and the zero induced middle map, gives

$$H_2(B\,\mathrm{El}_A\, H; \mathbb{Z}) \cong \ker(\mathrm{colim}_A K \xrightarrow{\mathrm{aug}} \mathbb{Z}).$$

This is the binary instance of the formula whose proof is given in TOTEX Part B; we checked it by these two short exact sequences and by the identification of the simplicial replacement of $\mathbb{Z}[H]$ with the nerve chain complex of $\mathrm{El}_A H$. These identifications are natural under the stated extension, since the old representables and their relation map are its restrictions. The augmentation of an element in the kernel of the old-to-new map on $\operatorname{colim} K$ is necessarily zero. Hence that whole kernel is precisely the kernel on H_2. Applying the first part with $F' = K$ proves the criterion. □

Example C.6 (Refuted literal wording: without monomorphisms the component need not be a circle)**.** The unqualified phrase "the component is a circle" in Lead 7 is false if it is read without the monomorphism hypothesis of the AKR application. Let A have objects x, T, S, arrows

$$A(x,T) = \{c,d\}, \qquad A(T,S) = \{f,g\}, \qquad A(x,S) = \{h\},$$

identity endomorphisms only, no other arrows, and the four compositions $fc = fd = gc = gd = h$. Composition with identities is prescribed in the usual way. This is a complete finite category: there are no three composable nonidentity arrows, and the identity laws settle every remaining associativity equation.

Take the five-vertex cyclic quasi-filter indexing graph ($k = 1$). It has copies S_i of S and copies T_i of T, with two arrows from T_i to S_i, S_{i+1} (cyclic indices), mapped by u to (g, f) for $i = 0, 1$, to (g, g) for $i = 2$, to (f, g) for $i = 3$, and to (f, f) for $i = 4$. The comma category $x \downarrow u$ has five S-vertices (S_i, h) and ten T-vertices $(T_i, c), (T_i, d)$, and twenty nonidentity arrows. It is connected. Its nerve is a graph, with $H_1 \cong \mathbb{Z}^{20-15+1} = \mathbb{Z}^6$; each S-vertex has valence four. Thus its component is neither isomorphic nor homotopy equivalent to a circle, although $fc = gc$ and its noncontractibility is exactly as asserted in the corrected proposition. The complete composition table, all typed composable triples, and these graph counts are checked by `akr_nonmonic_comma_circle.py` and its supplied output.

Proposition C.7 (Refuted: pointwise free split injections do not suffice for acyclicity)**.** *There exists a small nonempty connected quasi-filtered category A and a covariant functor $Q : A \to \mathrm{Ab}$ such that every $Q(X)$ is finitely generated free abelian, every $Q(g)$ is split injective with free abelian cokernel, and nevertheless*

$$T_1(Q) \cong \mathbb{Z}, \qquad T_i(Q) = 0 \quad (i \geq 2).$$

One may take any repaired AKR category $A = K^*$ and

$$Q = \operatorname{coker}\big(D : P_W \longrightarrow P_V^2\big), \qquad D(t) = \big(t(v_2 - v_1), t(v_3 - v_1)\big).$$

Thus pointwise freeness and split injectivity do not by themselves close the cycle-module colimit obstruction.

Proof. Proposition C.4 says that D is injective, so $0 \to P_W \xrightarrow{D} P_V^2 \to Q \to 0$ is a projective resolution of length one. Its induced colimit map is the zero map $\mathbb{Z} \to \mathbb{Z}^2$, because each component is a difference of two augmentation-one representable maps. Therefore $T_1(Q) = \ker(0 : \mathbb{Z} \to \mathbb{Z}^2) = \mathbb{Z}$ and all higher $T_i(Q)$ vanish.

We verify the asserted pointwise and transition properties integrally. For an arrow $g : Y \to X$, identify the images of $P_V(Y)$ and $P_W(Y)$ inside their counterparts at X, using monicity. The induced relative map

$$\overline{D} : P_W(X)/gP_W(Y) \longrightarrow P_V(X)^2/gP_V(Y)^2$$

has one column for every W-arrow $h : W \to X$ not factoring through g. By AKR C.2(c_m), its three-point image meets $g(Y)$ in at most one point. For any nonempty collection of these

columns, C.2(d_m) supplies more than $3r/2$ points outside $g(Y)$, where r is the number of columns. Hence one such outside point is private to one column. At that point the column has coordinate pair $(-1,-1)$, $(1,0)$ or $(0,1)$, according to its position in the triple. Choose a unit entry from that pair as pivot. Its row has zero entries in all other columns. Eliminating this column therefore leaves all other columns unchanged. Repeating gives a full unit-pivot reduction. All these matrices are finite because each object's incoming hom-sets are frozen at its finite stage. Thus $\overline{D}$ is injective and its cokernel is free abelian. The same argument without the relative quotient proves that $Q(X)$ is free abelian.

The exact diagram comparing the two short exact sequences at Y and X now gives, by the kernel–cokernel sequence,

$$0 \longrightarrow Q(Y) \xrightarrow{Q(g)} Q(X) \longrightarrow \operatorname{coker} \overline{D} \longrightarrow 0.$$

The final group is free, so this sequence splits. This verifies every claimed transition property while retaining the nonzero derived colimit calculated above. □

Corollary C.8 (Proved: the same obstruction with free groups and free-factor maps)**.** *For the same quasi-filtered category $A = K^*$ there is a covariant functor $\Gamma : A \to \mathrm{Grp}$ such that every $\Gamma(X)$ is a finitely generated free group and every transition map identifies its source with a free factor of its target, but the abelianization functor $\Gamma_{\mathrm{ab}} : A \to \mathrm{Ab}$ has $T_1(\Gamma_{\mathrm{ab}}) \cong \mathbb{Z}$.*

Proof. For each X, take group generators $x_{v,1}, x_{v,2}$ indexed by $v \in A(V,X)$ and impose, for each $h \in A(W,X)$, the relation

$$x_{hv_2,1}\, x_{hv_1,1}^{-1}\, x_{hv_3,2}\, x_{hv_1,2}^{-1} = 1.$$

Postcomposition on v and h defines a homomorphism on these presentations for every arrow of A, and makes Γ a covariant functor. Its abelianization is exactly the functor Q of Proposition C.7.

For any nonempty family of remaining relations the private-vertex argument supplies a vertex occurring in just one of them. One of the two generators indexed by that vertex occurs exactly once in that relation and nowhere in the others. At hv_1 we may choose the coordinate-one generator; the coordinate-two occurrence is a different generator. A Tietze elimination solves for the chosen generator and removes that sole relation, leaving all other relations unchanged. Repeating removes every relation and proves that $\Gamma(X)$ is free.

For an arrow $g : Y \to X$, retain the entire old presentation at Y. Apply the relative private-vertex argument only to the new relations, those indexed by W-arrows not factoring through g. Each chosen private vertex lies outside the old image, so the eliminated generator is new and occurs in no old relation. Eliminating all new relations leaves the old presentation together with additional free generators. Thus the natural map identifies $\Gamma(Y)$ with a free factor of $\Gamma(X)$. No compatible choices of free bases or of retractions are required. Finally the preceding proposition computes the stated nonzero derived colimit of its abelianization. □

Proposition C.9 (Proved: a free-factor nerve-colimit obstruction)**.** *There is a diagram of finitely generated free groups, with every transition a free-factor inclusion and with free group colimit, whose ordinary simplicial-set colimit of nerves has second integral homology $\mathbb{Z}$.*

Proof. Use the category consisting of a parallel pair $f, g : 0 \rightrightarrows 1$ and the diagram

$$G_0 = F(a,c), \qquad G_1 = F(a,c,b), \qquad f = (a,c), \quad g = (ab, cb^2).$$

The automorphism $a \mapsto ab$, $c \mapsto cb^2$, $b \mapsto b$ has inverse $a \mapsto ab^{-1}$, $c \mapsto cb^{-2}$, $b \mapsto b$. Thus both maps have free-factor image.

We claim that their image subgroups intersect only in the identity. For a nonempty reduced word w in a, c, consider $g(w)$ in the free product $F(a,c) * F(b)$. A negative first letter

gives a nonzero initial b-block, and a positive last letter gives a nonzero terminal b-block. In the remaining case, w starts positively and ends negatively, so it has a positive-to-negative transition. At such a transition the two underlying generators differ, since w is reduced. Its b-exponents are therefore $1-2$ or $2-1$, both nonzero. These are the only places where b-blocks from distinct letters can merge. No intervening a, c block reduces to the identity: it is a subword of the original reduced word. Hence at least one nonzero b-block survives, and $g(w) \notin F(a,c)$. This proves the claim for arbitrary word length.

Consequently $BG_0 \vee BG_0 \to BG_1$, induced by f, g, is a simplicial monomorphism. Indeed its two images intersect in each nerve degree only in the all-identity simplex. The ordinary coequalizer Y of Bf, Bg is the pushout of this monomorphism along the fold $BG_0 \vee BG_0 \to BG_0$. This is a homotopy pushout: realizations of simplicial monomorphisms are inclusions of CW subcomplexes, or one may factor the fold through its mapping cylinder before pushing out.

Free groups have classifying spaces homotopy equivalent to graphs. The Mayer–Vietoris sequence therefore gives

$$H_2(Y;\mathbb{Z}) \cong \ker(f_{\mathrm{ab}} - g_{\mathrm{ab}} : \mathbb{Z}^2 \to \mathbb{Z}^3) = \mathbb{Z}(2a - c),$$

since $(f_{\mathrm{ab}} - g_{\mathrm{ab}})(a) = -b$ and its value on c is $-2b$. On the other hand,

$$\operatorname{colim} G = \langle a, c, b \mid a = ab,\ c = cb^2 \rangle \cong F(a,c).$$

The first relation kills b, and the second is redundant. Thus the comparison $Y \to B(\operatorname{colim} G)$ cannot be a weak equivalence. □

Scope of the obstruction. The parallel-pair indexing category is not quasi-filtered and is not total. This example does not refute totality-to-exactness or establish anything about the totality of AKR. It rules out deriving asphericity of an ordinary nerve colimit solely from free values, free-factor transitions and a free group colimit.

The program `finite_freefactor_nerve.py` checks the Nielsen inverses, exact word identities certifying the group quotient, and the integer Mayer–Vietoris matrix. The latter has a unit 3×3 minor, all 4×4 minors zero, and primitive kernel generator $(2, -1, -2, 1)$. Its additional check of all 39,364 reduced nonempty words of length at most nine is supplementary; the proof above is unbounded.

C.2 An exact fixed-path test for a repaired attachment

The following statement strengthens the cycle-space criterion of Proposition C.5. It concerns the repaired path construction, with its prescribed target covered by the images of the generating maps. Relabelling that target does not affect the statement. In particular it applies with every admissible numerical choice in that repaired construction.

Proposition C.10 (Proved: the kernel is generated by fixed-endpoint paths)**.** *Let A be a finite stage, or a full finite substage containing V, W and every prescribed incoming generator and factorization for its retained targets, of the repaired AKR construction. Adjoin a single quasi-filter target Z for $f, g : S \rightrightarrows T$ and write A' for the resulting category. Put $s = 4k+1$, where $k \geq n$, and write $\alpha_i : T \to Z$, $i \in \mathbb{Z}/s$, for its generating maps. All old objects and arrows remain unchanged. For an object X let G_X have vertices $A(V,X)$ and one directed edge b from bv_1 to bv_2 for each $b \in A(W,X)$, using A' when $X = Z$. Let $K(X) = H_1(G_X;\mathbb{Z})$ and $Q = \operatorname{colim}_A K$.*

Let $F \subseteq G_S$ be the equalizer subgraph: its vertices are those $c : V \to S$ with $fc = gc$, and its edges are those $b : W \to S$ with $fb = gb$. Then

$$\ker(Q \longrightarrow \operatorname{colim}_{A'} K) = \langle [(g_\# - f_\#)\sigma] \ :\ \sigma \in C_1(G_S;\mathbb{Z}),\quad \partial\sigma = w - v,\quad v, w \in F_0 \rangle\,. \quad \text{(FP)}$$

Only pairs v, w lying in the same component of G_S occur. A path may be chosen for σ; reversing edges is represented by negative coefficients. The displayed difference is a cycle in G_T, and its class in Q is independent of the chosen path. It is enough to choose one vertex from each component of F and paths joining those representatives which lie in a common component of G_S.

Consequently this attachment preserves Q, and hence the old second homology of the binary category of elements, if

$$\pi_0(F) \longrightarrow \pi_0(G_S) \quad \textit{is injective.} \tag{FC}$$

More generally, preservation holds if and only if all classes in (FP) are zero in Q; injectivity in (FC) is sufficient but is not asserted to be necessary.

Proof. We first identify the cell fibers of the attachment. Every underlying point of every prescribed repaired target is the value of a V-arrow. This holds for V and W. It is preserved by a disjoint-sum target. For a quasi-filter target it follows because its fixed points, private points, and each point of every repaired path target are covered by the generating copies: the interval formulas in Proposition C.1 of the supplied corrected construction give a nonempty visibility interval for every path-target point. Induction gives the assertion. Thus the vertices of each G_X may be identified with its underlying points. No assertion about arbitrary extra, unused points adjoined to a target is needed here.

For a point $z \in Z$, put $I_z = \{i : z \in \alpha_i(T)\} \subseteq \mathbb{Z}/s$. Property (10) makes I_z geodesically convex in the odd index cycle. Indeed, between any two indices there is a unique shortest cyclic arc of length at most $2k \leq s - 2n$. Starting at either end, (10) puts the next index of that arc in I_z, and iteration puts the entire arc in I_z. A nonempty proper connected subset of a cycle is an interval, so I_z is either an interval or the whole cycle.

For an edge $e : W \to Z$, its visibility set is

$$I_e = \{i : e = \alpha_i b \text{ for some } b : W \to T\} = \bigcap_{r=1}^{3} I_{ev_r}.$$

The nontrivial inclusion follows from the established AKR routing invariant (c_{m+1}): if $e(W) \subseteq \alpha_i(T)$, its three image points, in particular at least two, lie in that image, so e factors through α_i. Each factorization is unique because α_i is injective. Intersections of geodesically convex sets are geodesically convex. Consequently each nonempty I_e is also an interval or the whole cycle.

Define the neighboring legs $\lambda_i, \mu_i : S \to T$ by

$$(\lambda_i, \mu_i) = \begin{cases} (g, f), & 0 \leq i < 2k, \\ (g, g), & i = 2k, \\ (f, g), & 2k < i < 4k, \\ (f, f), & i = 4k. \end{cases}$$

The repaired identities say $\alpha_i \lambda_i = \alpha_{i+1} \mu_i$. Property (11) says that the image of this composite is exactly $\alpha_i(T) \cap \alpha_{i+1}(T)$. Therefore the fiber graph of a vertex z has one vertex at each index of I_z and one edge between each neighboring pair of such indices. For an edge e, the same assertion follows by one further application of the old routing invariant: the unique lift $b : W \to T$ in copy i has its image in $\lambda_i(S)$ whenever $i, i+1 \in I_e$, so it factors uniquely through λ_i. Its factor $d : W \to S$ gives the neighbor lift $\mu_i d$ in copy $i+1$. Conversely every such source edge gives this neighbor relation. Thus, in either cell dimension, every fiber graph is a path or a circle, including a one-vertex path.

The circular fibers are precisely the equalizing cells. Here is the explicit verification, which also identifies their generators. Suppose a cell lies in all copies and let d_i be its source lift on

the neighboring overlap at i. Its lift in copy i is simultaneously $\lambda_i d_i = \mu_{i-1} d_{i-1}$, with cyclic indices. The two legs are injective on vertices and on edges. At the upper junction these equalities imply $d_{2k+1} = d_{2k-1}$, and successive cancellation gives

$$d_{2k+j} = d_{2k-j} \qquad (1 \le j \le 2k).$$

In particular $d_{4k} = d_0$. The lower junction then gives $g d_0 = f d_{4k} = f d_0$. Working forward through the first half, and then through the second half, forces every $d_i = d_0$. Conversely an equalizing cell d gives a circular fiber with every $d_i = d$.

Write $C_*(G)$ for the cellular chain complex of an oriented graph, concentrated in degrees zero and one. Define a chain map

$$\delta : C_*(G_S)^s \longrightarrow C_*(G_T)^s, \qquad (\delta x)_i = (\lambda_i)_\# x_i - (\mu_{i-1})_\# x_{i-1},$$

and define $\epsilon : C_*(G_T)^s \to C_*(G_Z)$ by $\epsilon(y) = \sum_i (\alpha_i)_\# y_i$. The cell-fiber graphs just computed show, integrally and degree by degree, that

$$0 \longrightarrow C_*(F) \xrightarrow{j} C_*(G_S)^s \xrightarrow{\delta} C_*(G_T)^s \xrightarrow{\epsilon} C_*(G_Z) \longrightarrow 0 \tag{CF}$$

is exact, where j is diagonal inclusion. For a path fiber the incidence map has zero kernel and cokernel $\mathbb{Z}$; for a circle fiber its kernel is the diagonal $\mathbb{Z}$ and its cokernel is $\mathbb{Z}$. These descriptions use bases and hence introduce no torsion. They commute with the boundary because all displayed maps are graph maps.

For completeness, F is the subgraph induced on its fixed vertices. If the endpoints of b are fixed, then fb and gb share their first two image points. Invariant (c_m) factors one through the other. The only endomorphism of the initial object W is its identity, since the construction adds only new maximal targets and leaves old hom-sets unchanged. Hence $fb = gb$. This induced-subgraph description is useful for testing (FC), but the chain argument only uses the equalizer definition.

Let $D_* = \operatorname{im} \delta$ be the image chain complex in $C_*(G_T)^s$. Splitting (CF) into two short exact sequences of complexes gives

$$\ker\big(K(T)^s \longrightarrow K(Z)\big) = \operatorname{im}\big(H_1(D_*) \longrightarrow K(T)^s\big),$$

and

$$K(S)^s \longrightarrow H_1(D_*) \longrightarrow H_0(F) \xrightarrow{H_0(j)} H_0(G_S)^s. \tag{HC}$$

The last map is diagonal on the component-inclusion homomorphism. Its kernel is therefore exactly $\ker(H_0(F) \to H_0(G_S))$. That kernel is generated by differences $[w] - [v]$ of components of F lying in one component of G_S.

Choose a graph path σ from v to w. Its diagonal copy in $C_1(G_S)^s$ has boundary $j(w - v)$, so $\delta(\sigma, \dots, \sigma)$ is a cycle in D_* lifting that component difference, up to the immaterial common boundary sign. Its ith component

$$\kappa_i = (\lambda_i)_\# \sigma - (\mu_{i-1})_\# \sigma$$

is separately a cycle in G_T, because f and g agree at v, w. The sum of these components is

$$\sum_i \kappa_i = \sum_i ((\lambda_i)_\# - (\mu_i)_\#)\sigma = (g_\# - f_\#)\sigma.$$

Here there are $2k$ occurrences of (g, f), $2k-1$ of (f, g), and two zero differences. On the other hand, the image in Q of every element coming from $K(S)^s$ in (HC) is zero: each of its terms is the difference of the images of a source cycle along two old arrows. Proposition C.5 now gives exactly (FP), in both directions.

If σ' has the same boundary as σ, their difference is a cycle $\tau \in K(S)$, and $[(g_\# - f_\#)\tau] = [\tau] - [\tau] = 0$ in Q. Moving either endpoint within its component of F also leaves the class unchanged, since $f_\# = g_\#$ on chains of F. These observations establish all path and representative assertions. Finally, the dimension-shifting formula identifies the old second homology with the kernel of the augmentation $Q \to \mathbb{Z}$; every generator in (FP) has augmentation zero. Thus the criterion for preserving Q is also the criterion for preserving that old second homology. □

Exact remaining issue. Proposition C.10 does not establish (FC) for arbitrary stages. The finite tests of that condition therefore address the exact possible attachment kernel, but passing those tests does not prove its eventual vanishing. Conversely a disconnected fixed subgraph alone would not prove a nonzero kernel: one must compute the associated class $[(g_\# - f_\#)\sigma]$ in the old colimit. Even a nonzero kernel would only show that some old second-homology class dies; it would not by itself show that the full infinite category is total or that its binary second homology vanishes.

Lemma C.11 (Proved: exact fixed subsets of the repaired copy maps)**.** *Consider a repaired attachment for injections $f, g : S \rightrightarrows T$ between finite sets. Assume that the partial relation $f(x) \mapsto g(x)$ has no nontrivial directed cycles. Let $F = \{f(x) : f(x) = g(x)\}$ and write the nonconstant directed paths of the partial relation $f(x) \mapsto g(x)$ as*

$$p_{\nu,0} \longrightarrow \cdots \longrightarrow p_{\nu,l_\nu - 1}.$$

Here l_ν is the number of vertices of the path, not its number of edges. Put $L = \max_\nu l_\nu$, or $L = 1$ when there are no such paths. For $s = 4k + 1$ and $k \geq n \geq L$, let $\alpha_0, \ldots, \alpha_{4k} : T \to Z$ be the repaired copy maps. For $0 \leq i < j \leq 4k$, their set of fixed source vertices is exactly:

1. *if $(i, j) = (r, 4k - r)$ with $0 \leq r \leq L - 2$, it is*

$$F \ \cup \ \{p_{\nu,t} : 0 \leq t \leq l_\nu - r - 2\};$$

2. *if $(i, j) = (2k - r, 2k + 1 + r)$ with $0 \leq r \leq L - 2$, it is*

$$F \ \cup \ \{p_{\nu,t} : r + 1 \leq t \leq l_\nu - 1\};$$

3. *for all other pairs it is F.*

An empty index range contributes no points. Thus at most $2L - 1$ distinct fixed subsets have to be tested, independently of s.

Proof. Fixed points in F are shared by every copy; private points are shared by no two distinct copies. On a path of length l, write t for its zero-based position. Its repaired coordinate is

$$\beta_i(t) = \begin{cases} t + i, & i \leq 2k,\ t + i \leq 2k, \\ t + i + 2k - l + 1, & i \leq 2k,\ t + i > 2k, \\ t - i + 6k - l + 2, & i > 2k,\ t - i + 6k - l + 2 > 2k, \\ t - i + 4k, & i > 2k,\ t - i + 6k - l + 2 \leq 2k. \end{cases}$$

For fixed t, this expression is strictly increasing on the first arm and strictly decreasing on the second arm: within each branch its step is respectively 1 or -1, and at a branch transition the jump has the same sign because $l \leq k$. Thus an equality between distinct copies must use $i \leq 2k < j$.

If both values lie in the lower branch, equality says $i + j = 4k$. Writing $i = r$, the branch inequalities reduce exactly to $t \le l - r - 2$. If both lie in the upper branch, equality says $i + j = 4k + 1$. Writing $i = 2k - r$, the inequalities reduce to $t \ge r + 1$. For the first-lower/second-upper case, equality would give $i + j = 6k - l + 2$; the lower inequality forces $j \ge t + 4k - l + 2$, contrary to the upper inequality. For the first-upper/second-lower case, equality would give $i + j = 2k + l - 1$, whereas the lower inequality implies $j \ge t + 4k - l + 2$, impossible since $l \le k$. These calculations give precisely the two displayed families, path by path. Different paths have different tags in the repaired target. Only $l \le k$ was used, so the calculation applies to every admissible common bound n and every actual path size $l \le n \le k$; no restriction to the tested size-three examples is involved. □

Corollary C.12 (Proved: the small-overlap case is harmless)**.** *Assume that f and g induce injections on binary graph components and that their underlying point images meet in at most one point. Then every equalizer subgraph of a pair of copy maps $\alpha_i, \alpha_j : T \to Z$ is connected in each component of G_T. Thus the fixed-path criterion applies to a subsequent attachment for any such copy pair, even when $K(T)/F(T)$ is nonzero.*

Proof. An internal vertex of a nonconstant point path belongs to $f(S) \cap g(S)$. Hence every path has length at most three, at most one has length three, and a fixed point cannot coexist with a length-three path. In Lemma C.11, the first prefix and suffix are respectively $f(S)$ and $g(S)$, including the fixed points. All remaining subsets have at most one point. The image of each source graph component is connected, and injection on components prevents two distinct source components from lying in one ambient component. This proves the claim for the two full images; the singleton and empty cases are immediate. The equalizer graph is induced on its fixed vertices by the established AKR two-point routing property, as in Proposition C.10. □

Theorem C.13 (Proved: protected first-stage cycles and a split-monic quotient)**.** *In any repaired AKR Construction C.1, let P_+ and P_- be the fresh first-stage quasi-filter targets for (v_1, v_2) and (v_2, v_1). Write s_+, s_- for their respective numbers of quasi-filter maps. In the binary graph $G_{P_\pm}$, let $S_\pm$ be the $s_\pm$ vertices of its unique circular component, and choose its integral cycle $\kappa_\pm$ with augmentation one. For $X \in K^*$ define*

$$F(X) = \mathbb{Z}[K^*(P_+, X)] \oplus \mathbb{Z}[K^*(P_-, X)], \qquad \iota_X(b) = b\kappa_\pm \in K(X) \quad (b : P_\pm \to X).$$

Then $\iota : F \to K$ is a natural monomorphism. Its quotient $M = K/F$ is pointwise a finitely generated free abelian group, and every arrow of K^* induces a split injection on M with free abelian cokernel. The splitting is an existence statement for each individual arrow; no compatible family of splittings is asserted.

More precisely, if $X \in K_m$, $m \ge 1$, $g : Y \to X$ is any arrow, and $h_i : P_{\epsilon_i} \to X$ is a finite family of distinct typed arrows, then:

1. *$|h_i(S_{\epsilon_i}) \cap g(Y)| \ge 3$ implies that h_i factors through g;*
2. *if every displayed intersection has at most two points, then*

$$\left| \bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y) \right| \ge \frac{a_m}{a_1} \sum_i (s_{\epsilon_i} - 2).$$

These assertions hold for mixed families with either source P_+ or P_-.

Proof. We first verify the marked-set versions of the source induction. At stage one, a map from P_+ or P_- to an old object is its identity, if the target is that same object, and does not otherwise exist. Every proper incoming map to $P_\pm$ has source V or W. Its image meets the

marked circle in at most two points: a W-map supplies one edge of that circle and its third point lies outside it. Therefore the first assertion holds at stage one, and the second has bound $s_{\pm} - 2$ for the only possible nonempty family. There are no arrows between P_+ and P_-, and the only endomorphisms of either are identities; this remains so because all stages are full extensions.

We give the quasi-filter step explicitly. Write the old pair as $\phi_1, \phi_2 : R \rightrightarrows T$ and its new maps as $\alpha_j : T \to Z$, with $j \in \mathbb{Z}/s$. An identity comparison map $g = 1_Z$ makes the first assertion trivial and the second vacuous. Otherwise write $g = \alpha_0 g^*$ after a cyclic renumbering. Every marked arrow into Z factors as $h = \alpha_j h^*$.

Suppose first that $h(S_\epsilon)$ meets $g(Y)$ in at least three points. If $1 \le j \le s - 2n$, (10) gives $\alpha_j(T) \cap \alpha_0(T) \subseteq \alpha_j(T) \cap \alpha_{j-1}(T)$. If $s - 2n < j < s$, move instead toward $j + 1$; its forward distance to 0 is less than $2n \le s - 2n$, so the other containment in (10) applies. In either case choose the adjacent identity

$$\alpha_j \phi_p = \alpha_{j'} \phi_q, \qquad j' = j - 1 \text{ or } j + 1, \quad p, q \in \{1, 2\}.$$

By (11) the intersection of the two copies is exactly $\alpha_j \phi_p(R)$. Injectivity of α_j therefore gives $|h^*(S_\epsilon) \cap \phi_p(R)| \ge 3$. The old first assertion supplies $l : P_\epsilon \to R$ with $h^* = \phi_p l$. Hence $h = \alpha_{j'} \phi_q l$: the whole typed arrow, not merely its marked set, has moved to the adjacent copy. Its image and its marked image are unchanged. If we started with $j \le s - 2n$, repeat downward to 0; otherwise repeat upward through $s - 1$ to 0. Each relevant distance decreases, so (10) remains applicable. Once $h = \alpha_0 h_0^*$, injectivity of α_0 and the old first assertion applied to h_0^*, g^* show that h factors through g. This proves the new first assertion without any use of the number of points of P_ϵ.

For the second assertion let the family be $h_1, \dots, h_r$, give member i the positive weight $w_i = s_{\epsilon_i} - 2$, and put $\beta_m = a_m / a_1$. Define disjoint sets

$$H_j = \{i : h_i \text{ factors through } \alpha_j\} \setminus \bigcup_{t<j} H_t, \qquad 0 \le j < s.$$

They partition the family. Partition the indices $0, \dots, s - 1$ into $s/(3n)$ consecutive blocks of length $3n$. Choose a block $B = \{j_0, \dots, j_0 + 3n - 1\}$ minimizing $\sum_{j \in B} \sum_{i \in H_j} w_i$, and discard its assigned family members. For the retained index set $L = \{1, \dots, r\} \setminus \bigcup_{j \in B} H_j$ we have

$$\sum_{i \notin L} w_i \le \frac{3n}{s} \sum_{i=1}^{r} w_i, \qquad \sum_{i \in L} w_i \ge \frac{a_{m+1}}{a_m} \sum_{i=1}^{r} w_i.$$

Now order the remaining copy indices by

$$0, 1, \dots, j_0 - 1, \quad s - 1, s - 2, \dots, j_0 + 3n,$$

omitting the first arm if $j_0 = 0$. Assign each retained member to the first available copy in this order. Denote by L_j the family assigned to j, write $h_i = \alpha_j h_i^*$ for $i \in L_j$, and put $D_j = \bigcup_{i \in L_j} h_i(S_{\epsilon_i})$.

If the first index is 0, the old second assertion, applied to g^* and the h_i^* assigned there, yields

$$|D_0 \setminus g(Y)| \ge \beta_m \sum_{i \in L_0} w_i.$$

For every later index j in the first arm, all intersections with earlier copies and with $g(Y)$ lie, by (10), in $\alpha_j(T) \cap \alpha_{j-1}(T)$. Indeed all backward distances are at most $j_0 - 1 < s - 2n$. For an index j in the second arm, every earlier index, and also 0, is reached in the forward cyclic direction in at most $s - 3n$ steps, so those intersections lie in $\alpha_j(T) \cap \alpha_{j+1}(T)$. This includes the first index $s - 1$ when $j_0 = 0$.

Let j' be this adjacent comparison index. No member of L_j factors through $\alpha_{j'}$. Usually j' is earlier in the sweep, so this follows from the assignment rule. In the only exceptional case $j = s-1, j_0 = 0$, a factorization through α_0 would put that member in H_0, hence in the discarded block, again impossible. Choose $\alpha_j\phi_p = \alpha_{j'}\phi_q$ as above. If $h_i^*(S_{\epsilon_i})$ met $\phi_p(R)$ in three points, the old first assertion would factor h_i^* through ϕ_p and hence h_i through $\alpha_{j'}$, a contradiction. Therefore the old second assertion applies to the family h_i^* and comparison arrow ϕ_p, and (11) gives

$$\begin{aligned}\left|D_j \setminus \left(g(Y) \cup \bigcup_{t \text{ earlier}} D_t\right)\right| &\geq |D_j \setminus \alpha_j\phi_p(R)| \\ &= \left|\bigcup_{i\in L_j} h_i^*(S_{\epsilon_i}) \setminus \phi_p(R)\right| \\ &\geq \beta_m \sum_{i\in L_j} w_i.\end{aligned}$$

Summing these disjoint increments over the sweep and using the retained-weight estimate proves

$$\left|\bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y)\right| \geq \beta_m \sum_{i\in L} w_i \geq \beta_{m+1} \sum_{i=1}^{r} w_i.$$

Each factorization preserves the source and its weight; injectivity of a copy map preserves distinctness of the typed arrows. Thus the argument is valid for mixed protected source types. It uses only (10), (11), monicity, the old threshold-three assertion, and the old weighted bound. The original special cardinality $|W| = 3$ is not used in this inductive step.

At a disjoint-sum target, the two summands have disjoint point sets. If the comparison arrow is the identity of the new target, the factorization assertion is immediate and the nonfactorization bound is vacuous. Otherwise every nonempty marked overlap places the marked arrow and the comparison arrow in the same summand; the old assertion then gives the required factorization. Apply the old bound in the comparison summand. In the other summand one may use the following unconditional bound. Choose one marked arrow h_r as comparison. Every other member has at most two marked points in its full image, since otherwise it would factor through h_r. Such a factorization between the two fixed source objects is impossible unless both sources and arrows coincide. Applying the old second assertion to the other members and then adding all s_{ϵ_r} points of $h_r(S_{\epsilon_r})$ gives at least $(a_m/a_1)\sum_i w_i$, because $(a_m/a_1)(s_{\epsilon_r} - 2) \leq s_{\epsilon_r}$. This proves both marked assertions for all stages.

Every admissible first-stage length satisfies $s_\pm \geq 45$. Indeed its active path has two vertices, divisibility $3n \mid s = 4k+1$ forces n odd, hence $n \geq 3$, and $s/(3n) \geq (1 - a_1/2)^{-1} > 4$ is an integer. Also $a_m/a_1 > 3/4$. Consequently

$$\frac{a_m}{a_1}(s_\pm - 2) > \frac{s_\pm}{2}.$$

The unconditional bound from the preceding paragraph shows that every nonempty finite family of marked circles has a vertex belonging to exactly one of them: otherwise the union would have at most half the total number of marked incidences. At such a vertex, an incident edge of that circle belongs to no other circle in the family. Its coefficient detects the coefficient of that circle in a proposed linear relation. This proves injectivity of ι_X, over $\mathbb{Z}$ and indeed with arbitrary abelian-group coefficients. Naturality follows from composition of arrows.

For the quotient assertion, attach a two-cell to G_X along each marked embedded circle $b\kappa_\pm$ and denote the resulting finite CW complex by C_X. The preceding private-vertex argument provides a boundary edge of one two-cell that belongs to no other two-cell. Collapsing that

cell across that edge and repeating removes all two-cells. Thus C_X is homotopy equivalent to a graph, and

$$H_1(C_X;\mathbb{Z}) = K(X)/F(X) = M(X)$$

is free abelian.

An arrow $g : Y \to X$ embeds C_Y cellularly into C_X: it is injective on vertices, edges and typed protected cells. The cells not in its image correspond exactly to protected arrows that do not factor through g. For any nonempty family of those cells the marked assertions give

$$\left|\bigcup_i h_i(S_{\epsilon_i}) \setminus g(Y)\right| > \frac{1}{2}\sum_i s_{\epsilon_i}.$$

Hence there is a private marked vertex outside $g(Y)$. A boundary edge incident to it is free among the new two-cells and is not an edge of $g(C_Y)$. We can therefore perform all collapses relative to $g(C_Y)$. The resulting relative CW complex has only additional zero- and one-cells. Adding such cells injects each old first homology group and has free abelian cokernel, as follows directly by extending a maximal forest. Thus $M(g)$ is injective with free cokernel and, over $\mathbb{Z}$, is split injective. □

The remaining colimit obstruction after this theorem. The theorem is pointwise and relative to individual arrows. It does not assert injectivity after taking a colimit. The long exact sequence of $0 \to F \to K \to M \to 0$ contains

$$T_1(M) \longrightarrow \mathbb{Z}^2 \longrightarrow \operatorname{colim}_{K^*} K.$$

The two standard basis vectors are the two protected cycles. Their augmentation difference $(1,-1)$ is the first-stage two-dimensional class. The theorem has not shown that this difference lies outside the image of $T_1(M)$.

C.3 Sharper intersections of the protected cycles

Proposition C.14 (Proved: a proper overlap lies on one boundary edge)**.** *Use the protected objects P_+, P_-, their marked circles S_+, S_-, and their cycle generators from Theorem C.13. For every object X of the repaired construction, every protected arrow $h : P_\epsilon \to X$, and every arrow $g : Y \to X$, the following stronger routing assertion holds. If $h(S_\epsilon) \cap g(Y)$ contains the images of two vertices which are not adjacent in S_ϵ, then h factors through g. Consequently, if h does not factor through g, that intersection is empty, a single vertex, or two adjacent vertices. The boundary graphs of two distinct protected cells in G_X meet in an empty set, one vertex, or one edge.*

Proof. We prove the routing assertion by stage induction. At the first stage there is no arrow between the two distinct protected objects and no nonidentity endomorphism of either. A proper incoming V-arrow meets the marked circle in at most one vertex; a W-arrow meets it in the endpoints of its one boundary edge. Thus the assertion holds at the birth stage.

Suppose first that X is a new quasi-filter target, with neighboring maps $\alpha_i : T \to X$ and old pair $\phi_0, \phi_1 : S \rightrightarrows T$. The case $g = 1_X$ is immediate. Otherwise cyclically rename the copies so that $g = \alpha_0 g^*$ and write $h = \alpha_j h^*$. Choose two nonadjacent marked vertices $p, q \in S_\epsilon$ whose images lie in $g(Y)$. If $j \neq 0$, take the next index j' on the shortest arc from j to 0. Its total length is at most $2k \leq s - 2n$, so (10) puts $h(p), h(q)$ in the overlap of copies j, j'. By (11) that overlap is $\alpha_j\phi_r(S) = \alpha_{j'}\phi_t(S)$ for appropriate $r, t \in \{0,1\}$. Injectivity of α_j puts $h^*(p), h^*(q)$ in $\phi_r(S)$. The old induction hypothesis, applied to the same two nonadjacent vertices in the fixed source P_ϵ, gives an old arrow $l : P_\epsilon \to S$ with $h^* = \phi_r l$. Hence $h = \alpha_{j'}\phi_t l$. The entire protected arrow has moved one step towards copy zero. Repeating at most $2k$ times writes

$h = \alpha_0 \widetilde{h}$. Injectivity then puts $\widetilde{h}(p), \widetilde{h}(q)$ in $g^*(Y)$, and the old assertion factors $\widetilde{h}$ through g^*. This factors h through g.

At a disjoint-sum target, $g = 1_X$ is again immediate. Otherwise any nonempty overlap puts h and g in the same summand, where the old assertion applies. This completes the induction for every stage and every numerical choice. A subset of a cycle of length at least four with no nonadjacent pair has at most two vertices, and a two-vertex such subset is an edge's pair of endpoints.

Finally suppose two distinct protected arrows have two common marked vertices. If these were nonadjacent in either source circle, the assertion just proved would factor that protected arrow through the other. There are no arrows between P_+ and P_-, and their only endomorphisms are identities; such a factorization would therefore make the arrows equal. Thus the common vertices must be adjacent in both circles. Their two boundary W-arrows share two image points, so the original AKR routing invariant and $\mathrm{End}(W) = \{1\}$ make those W-arrows equal. A further common marked vertex would give a nonadjacent pair. The graph intersection is therefore exactly one edge in this case, proving the final assertion. □

Proposition C.15 (Proved: protected chains supported on two arrow images)**.** *Let $P(X) \subseteq K(X) \subseteq C_1(G_X; \mathbb{Z})$ be the subgroup freely generated by the protected cycles in Theorem C.13. For any two arrows $f : Y \to X$ and $g : Z \to X$, let $U \subseteq G_X$ be the union of their image subgraphs. Then*

$$P(X) \cap C_1(U; \mathbb{Z}) = f_{\#}P(Y) + g_{\#}P(Z). \tag{PI}$$

The same assertion holds with coefficients in an arbitrary abelian group. Thus a protected-cycle combination whose edge support lies in the union uses only protected cells factoring through one of those two arrows.

Proof. The inclusion from right to left follows from functoriality. For the other inclusion, use the unique expansion in the protected-cell basis, and consider the nonempty family, if any, of its cells which factor through neither f nor g. Index those cells by $i = 1, \dots, r$, write their marked images as S_i, and let $s_i = |S_i| \geq 45$. Each meets either full arrow image in at most two points, by the protected routing theorem. At a stage m containing all the data, put $q = a_m/a_1 > 3/4$. The weighted counting inequality gives

$$\left| \bigcup_i S_i \setminus f(Y) \right| \geq q \sum_i (s_i - 2).$$

Deleting the second image removes at most $2r$ further points. Consequently

$$\left| \bigcup_i S_i \setminus (f(Y) \cup g(Z)) \right| \geq q \sum_i (s_i - 2) - 2r > \frac{3}{4} \sum_i s_i - \frac{7}{2} r > \frac{1}{2} \sum_i s_i.$$

The final strict inequality uses $s_i \geq 45$. If every point of this outside union belonged to at least two of the indicated marked circles, the number of marked incidences would be at least twice that union's size, contradicting the bound by $\sum_i s_i$. There is therefore a marked vertex outside both images belonging to exactly one of the indicated cells. All other cells in the full expansion factor through f or g and have every marked vertex inside the union, so this vertex belongs to no other cell in the expansion.

Every boundary vertex of a protected cell is marked. Either boundary edge incident to this private vertex consequently belongs to that cell alone, and lies outside U. Its coefficient in the given combination is the nonzero cell coefficient, with sign ± 1. This contradicts the assumed edge support in U. Thus no such cell occurs, proving (PI). The argument only uses that changing sign preserves a nonzero coefficient, so it works for coefficients in any abelian group. □

Limitation of this local conclusion. Proposition C.15 is a statement about chains at one object. It does not assert that the protected subgroup remains injective after taking the colimit of K. A path-difference class from Proposition C.10 could agree with a protected combination only after old colimit relations have been used, rather than as chains at its target. Controlling those prior relations is still required for the all-stage assertion in C.1.

C.4 A convexity reduction for the exceptional AKR links

For a concrete AKR object X, let G_X be the graph whose vertices are the arrows $V \to X$ and whose edges are the arrows $b : W \to X$, joining bv_1 to bv_2. The functor $h_V/(bv_1 \sim bv_2)$ is $\pi_0 G_X$. In the repaired C.1 construction, the opened counting invariant implies that G_X is simple: distinct W-arrows cannot have the same two endpoints. It also implies that an arrow image induces the corresponding source graph, since a W-arrow whose two endpoints lie in another arrow image factors through that arrow.

Lemma C.16 (Proved: fixed sets of convex partial isometries)**.** *Let S, T be finite simple graphs and let $f, g : S \to T$ be induced graph embeddings. Suppose each connected component of each image is geodesically convex in T, and that both embeddings induce injections on connected components. Suppose also that the directed graph with edges $f(x) \to g(x)$ whenever $f(x) \neq g(x)$ has no directed cycle. Then the induced subgraph on*

$$F = \{x \in S : f(x) = g(x)\}$$

is geodesically convex within each connected component of S. In particular its intersection with every component of S is empty or connected.

Proof. Convex induced graph embeddings preserve distances within each connected component. Fix $u, v \in F$ in one component of S and put $a = f(u) = g(u)$, $b = f(v) = g(v)$. Let $I_S(u, v)$ be the finite set of all vertices lying on a shortest path from u to v, and define $I_T(a, b)$ similarly. Isometry and convexity give

$$f(I_S(u, v)) = I_T(a, b) = g(I_S(u, v)).$$

Thus $g^{-1}f$ is a permutation of the finite set $I_S(u, v)$. If it had a cycle of length greater than one, the corresponding values under g would give a nontrivial directed cycle of the relation $f(x) \to g(x)$, with the orientation reversed. This is excluded. Consequently this permutation is the identity, so $f(x) = g(x)$ for every $x \in I_S(u, v)$. Every shortest u–v path therefore lies in F, as required. □

Remark C.17 (Open in this incomplete record: convexity induction)**.** The repaired quasi-filter construction supplies the no-cycle hypothesis for its defining parallel pair. The lemma therefore reduces connectedness of its fixed-point intersections to convexity and component injectivity of the old arrows. Neither convexity preservation under every admissible quasi-filter attachment nor the resulting universal-choice persistence is proved here. Proposition C.10 supplies the exact cycle-space comparison, and Theorem C.24 completes the induction for a controlled choice of large targets. The remaining universal assertion must not be inferred from those choices or from the finite verifications below.

Proposition C.18 (Proved: a limitation of a generic preservation lemma)**.** *For arbitrary finite graphs, convex image components, injectivity on connected components, and bounded point-transition paths do not suffice to ensure that the repaired beta gluing is injective on connected components of each old copy. This assertion concerns a generic graph construction; the displayed arrow configuration is not asserted to occur in the category generated by C.1.*

Proof. Let S have two isolated vertices, let T have vertex set $\{0,1,2,3\}$ and edges $01, 23$, and define induced embeddings

$$f : S \to T, \quad f = (0,2), \qquad g : S \to T, \quad g = (3,1).$$

Each image component is a singleton, hence convex, and each map sends the two components of S to the two distinct components of T. The nontrivial point transitions are the disjoint paths $0 \to 3$ and $2 \to 1$.

Apply the actual-path-size beta formula to these two paths, with $k = 3$, $s = 13$, and the common bound $n = 3$. Write the two path-coordinate sets of the target as L_j, R_j for $0 \le j < 13$. If $(u_i, v_i) = (\beta_i^2(0), \beta_i^2(1))$, the image of the two edges of T in copy i is

$$L_{u_i} R_{v_i}, \qquad R_{u_i} L_{v_i}.$$

The thirteen unordered pairs $\{u_i, v_i\}$ are exactly the edges of the cycle

$$0, 1, 2, 3, 4, 5, 6, 12, 11, 10, 9, 8, 7, 0.$$

The resulting graph is therefore its bipartite double cover. Since the cycle has odd length 13, this double cover is a connected 26-cycle. In particular $L_0 = \alpha_0(0)$ and $R_0 = \alpha_0(2)$ have distance 13 in the new graph, although 0 and 2 lie in different components of T.

The file `finite_akr_graph_gluing.py` independently builds all thirteen maps, verifies that the target has 26 vertices and 26 edges, checks degree two and connectedness, and prints a length-13 path between these two vertices. The calculation uses the same repaired beta formula but makes no claim that the two displayed embeddings are among the arrows produced from the prescribed C.1 initial category. Thus it invalidates only the generic preservation lemma with these insufficient hypotheses. □

Additional exact finite checks. The program `finite_akr_intersections.py` reconstructs the selected second-stage attachment and the critical third-stage attachment specified in the earlier manuscript. It checks that every arrow induces an injection on $\pi_0 G$. For incoming images of source cardinality at least four it then computes the closure under all finite intersections and verifies that each intersection is connected within every ambient G-component that it meets. At the selected second-stage target there are 154 distinct images including the identity, 11781 unordered image pairs, and 307 distinct intersections with at least two vertices. At the critical third-stage target the corresponding counts are 892, 397386, and 1189. In both closures every new nontrivial intersection already occurs as an intersection of two generating images. No connectivity or component-injectivity failure occurs.

The separate program `finite_akr_convex.py` checks geodesic convexity directly. For every vertex of every nontrivial component of a proper incoming image it compares intrinsic graph distances with an ambient breadth-first search, also recording whether an equal-length shortest path can leave the image. It checks all 153 proper larger-source images at the selected second-stage target, using 6732 source searches, and all 891 such images at the critical third-stage target, using 52272 searches. All these image components are geodesically convex. The small-source image components in these finite stages are points or edges and so are geodesically convex automatically; component injectivity was checked separately for all arrows. The program would reconstruct an explicit shorter or equal-length path outside an image upon failure. None was found.

These checks concern the two stated finite substages, not complete later stages and not their infinite union. The output files include the exact object and arrow counts and the source-size restriction used in the intersection enumeration. They provide evidence for a possible convexity induction, not its proof.

Proposition C.19 (Refuted: the diameter condition cannot be dropped for generic connected graphs)**.** *For a generic pair of induced graph embeddings with connected source and target, strong convexity of both source images, injectivity on components, and bounded nontrivial point paths do not guarantee convexity of the old copies in the repaired beta target. This failure can occur at $n = 3$, $s = 45$, $k = 11$, even though these numbers satisfy the numerical C.1 condition for $a_0 = 2$, $a_1 = 8/5$. The graph configuration below is not asserted to arise among the arrows of the prescribed C.1 construction.*

Proof. More generally fix an odd $s = 4k + 1$ with $k \geq 2$, and an integer $L \geq s$. Let S be the path with vertices $0, \dots, L$, let T be the cycle with vertices $0, \dots, 2L + 1$ in cyclic order, and define

$$f(r) = r, \qquad g(r) = L + 1 + r \qquad (0 \leq r \leq L).$$

Both image paths have length L, strictly less than half the cycle length $L + 1$. Every shortest path in T between two vertices of either image is therefore the unique path in that image. Thus both embeddings are strongly convex. Source and target are connected, so their component maps are injective. The point-transition relation is the disjoint union of the two-vertex paths $r \mapsto L + 1 + r$. In particular any common bound $n \geq 2$ is valid. The two actual point maps are also increasing for the natural orders on the displayed vertex sets.

Use the actual-path-size beta construction. Its vertex set is $\{0, \dots, L\} \times \mathbb{Z}/s$, with copy i given by

$$\alpha_i f(r) = (r, u_i), \qquad \alpha_i g(r) = (r, v_i), \qquad (u_i, v_i) = (\beta_i^2(0), \beta_i^2(1)).$$

The graph has a vertical path of length L in every column, and for every index pair $\{u_i, v_i\}$ it has the two end bridges

$$(L, u_i)(0, v_i), \qquad (0, u_i)(L, v_i).$$

The unordered index pairs are the edges of the cycle

$$0, 1, \dots, 2k, 4k, 4k - 1, \dots, 2k + 1, 0.$$

Because s is odd, starting at $(0, 0)$ and following only the end bridges along this whole index cycle ends at $(L, 0)$ after s edges. The two vertices belong to copy zero, where their intrinsic distance is L. This bridge path leaves copy zero. If $L > s$, it is shorter than the old distance; if $L = s$, it has the old distance and is an outside shortest path. To verify the latter assertion, any path that uses an interior vertex of a vertical column and connects distinct column ends must traverse a full column and has length at least L. The bridge subgraph on the two ends of each column is the connected double cover of an odd s-cycle, hence a $2s$-cycle; the two ends of column zero are opposite and have bridge distance s. Thus the new distance is exactly $\min\{L, s\}$.

For a strict shortening at numerically admissible parameters, take $L = 46$, $n = 3$, $k = 11$, and $s = 45$. The source is a 46-edge path, the old target a 94-cycle, and the repaired target has 2115 vertices and 2160 edges. The indicated old distance 46 becomes 45. The standalone program `akr_connected_convex_counterexample.py` verifies both old convex embeddings, the point-path bound, all neighboring copy identities, the explicit 45-edge bridge path, and its exact new distance. Its JSON certificate specifies every map and every vertex of the shorter path.

The numerical compatibility is not a counterexample to universal C.1. In particular, if the displayed old cycle were one of the protected first-stage circles, either proper half-cycle arrow image would contradict Proposition C.14. The specific realization of such a configuration among later generated objects remains an additional requirement. □

C.5 Ordering a sufficiently long repaired necklace

We first specify the set-colimit assertion used in the ordering argument. The target in this subsection is exactly the covered repaired target: it has the prescribed fixed points, private points, and actual-length path targets. No additional unused points are required.

Lemma C.20 (Proved: the repaired target is the strict necklace colimit)**.** *Let $f, g : S \rightrightarrows T$ be injections of finite sets. Suppose the nonidentity part of the relation $f(x) \mapsto g(x)$ has only directed paths. Let $n \geq 2$ bound their numbers of vertices, choose $k \geq n$, and put $s = 4k + 1$. Let*

$$\alpha_i : T \longrightarrow Z_k, \qquad i \in \mathbb{Z}/s,$$

be the actual-path-length repaired beta maps. Define

$$(\lambda_i, \mu_i) = \begin{cases} (g, f), & 0 \leq i < 2k, \\ (g, g), & i = 2k, \\ (f, g), & 2k < i < 4k, \\ (f, f), & i = 4k. \end{cases}$$

Then Z_k is the coequalizer in sets of

$$\coprod_{i \in \mathbb{Z}/s} S \rightrightarrows \coprod_{i \in \mathbb{Z}/s} T, \qquad (i, x) \longmapsto (i, \lambda_i x), \quad (i, x) \longmapsto (i + 1, \mu_i x). \tag{NC}$$

In particular, compatible maps from the displayed copies of T to any set induce a unique map from Z_k.

Proof. Write a nontrivial relation path as $t_0 \mapsto t_1 \mapsto \cdots \mapsto t_{m-1}$, where $m \geq 2$. Its target consists of the s points $0, \ldots, 4k$, and the restriction of copy i is

$$\beta_i^m(t) = \begin{cases} t + i, & i \leq 2k,\ t + i \leq 2k, \\ t + i + 2k - m + 1, & i \leq 2k,\ t + i > 2k, \\ t - i + 6k - m + 2, & i > 2k,\ t - i + 6k - m + 2 > 2k, \\ t - i + 4k, & i > 2k,\ t - i + 6k - m + 2 \leq 2k. \end{cases}$$

Substitution gives $\alpha_i \lambda_i = \alpha_{i+1} \mu_i$. It also gives the exact consecutive overlap: for $i \leq 2k$ it is the image under β_i^m of $\{1, \ldots, m-1\}$, and for $i > 2k$ it is the image of $\{0, \ldots, m-2\}$. These are exactly the images of the appropriate source edges of the path under λ_i.

For clarity, every identification in the target is generated by these neighboring identifications. For a target point u, its visibility set $I_u = \{i : u \in \beta_i^m(\{0, \ldots, m-1\})\}$ is

range of u	I_u
$0 \leq u \leq m - 2$	$[4k - u, 4k] \cup [0, u]$
$m - 1 \leq u \leq 2k$	$[u - m + 1, u]$
$2k + 1 \leq u \leq 4k - m + 1$	$[6k - m + 2 - u, 6k + 1 - u]$
$4k - m + 2 \leq u \leq 4k$	$[u - 2k, 6k + 1 - u]$.

Each row is a nonempty cyclic interval. Because every β_i^m is injective, there is one preimage of u at each index of I_u. The consecutive-overlap calculation joins exactly the adjacent preimages. Thus every such fiber is connected by the relations in (NC).

A fixed point has one preimage in every copy, joined cyclically by those same relations. A private point has exactly one preimage. The three kinds of targets are disjoint, and the visibility table also proves that all path-target points are covered. Therefore $\coprod_i T \to Z_k$ is onto, and each of its fibers is exactly one equivalence class generated by the two displayed maps. This is the stated strict coequalizer. □

Proposition C.21 (Proved: all sufficiently long repaired necklaces are orderable)**.** *Let S, T be finite linearly ordered sets and let $f, g : S \rightrightarrows T$ be increasing injections. Fix $n \geq 2$ at least the number of vertices in every nontrivial path of the relation $f(x) \mapsto g(x)$. There is an integer $K \geq n$ such that, for every $k \geq K$, the covered repaired target Z_k admits a linear order making every generating map $\alpha_i : T \to Z_k$ increasing. The prescribed order on T is unchanged. The assertion permits any additional lower bounds on k.*

Proof. The partial bijection $f(S) \to g(S)$, $f(x) \mapsto g(x)$, is increasing. It has no nontrivial directed cycle: restriction to such a cycle would give a nonidentity increasing permutation of a finite linear order. Its nonfixed components are therefore paths, so Lemma C.20 applies.

Embed T increasingly in $\mathbb{R}$. Extend the increasing partial bijection to an increasing homeomorphism

$$h : \mathbb{R} \longrightarrow \mathbb{R}, \qquad h(f(x)) = g(x) \quad (x \in S). \tag{H}$$

For nonempty S, this can be done by linear interpolation through the finitely many points $(f(x), g(x))$, with slope-one translations on the two unbounded complementary intervals. If S is empty, take h to be the identity.

Every component J of $\mathbb{R} \setminus \mathrm{Fix}(h)$ is invariant under h. On J, either $h(t) > t$ everywhere or $h(t) < t$ everywhere. In the first case $h^r(t)$ tends to the right endpoint of J as $r \to \infty$; in the second it tends to the left endpoint. Indeed the iterates are monotone, and a finite limit in the interior would be fixed by continuity, contrary to the definition of J. The assertion includes convergence to an infinite endpoint.

For each $x \in S$, the points $f(x)$ and $g(x)$ either coincide in $\mathrm{Fix}(h)$ or lie in the same such component J. In the nonfixed case put

$$B_J = (f(S) \cup g(S)) \cap J.$$

Only finitely many components meet these finite sets. Consequently there is $K \geq n$ such that, for every $k \geq K$,

$$\begin{cases} h^{2k}(g(S) \cap J) > \max B_J, & h|_J > \mathrm{id}, \\ h^{2k}(g(S) \cap J) < \min B_J, & h|_J < \mathrm{id}\,. \end{cases} \tag{S}$$

An inequality for a finite set means that every element satisfies it. Once these inequalities hold, monotonicity of the iterates makes them hold for every larger k.

Fix such a k. There is an increasing homeomorphism w of $\mathbb{R}$ satisfying simultaneously

$$w(g(x)) = f(x) \quad (x \in S), \qquad w(h^{2k}g(x)) = h^{2k}g(x) \quad (x \in S). \tag{W}$$

To verify this, first consider the prescription on its finite domain. Within a positive component J, all points of $h^{2k}g(S) \cap J$ lie strictly above both $f(S) \cap J$ and $g(S) \cap J$; in a negative component they lie strictly below both. Thus the increasing bijection $g(x) \mapsto f(x)$ is compatible with fixing these additional points. Prescriptions in different components preserve their relative order. At fixed points the two prescriptions agree, since $g(x)$ fixed implies $f(x) = g(x)$ by injectivity of h. The finite prescription is therefore well-defined and strictly increasing. Linear interpolation, with slope-one tails, extends it to the required homeomorphism w.

Define increasing maps $a_i : T \to \mathbb{R}$ by

$$a_i = \begin{cases} h^i|_T, & 0 \leq i \leq 2k, \\ (w \circ h^{4k+1-i})|_T, & 2k < i \leq 4k. \end{cases}$$

They satisfy exactly the prescribed necklace equations. For $0 \leq i < 2k$, equation (H) gives

$$a_i g = h^i g = h^{i+1} f = a_{i+1} f.$$

At the upper junction, equation (W) gives

$$a_{2k}g = h^{2k}g = wh^{2k}g = a_{2k+1}g.$$

For $2k < i < 4k$,

$$a_i f = wh^{4k+1-i}f = wh^{4k-i}g = a_{i+1}g.$$

At the lower junction,

$$a_{4k}f = whf = wg = f = a_0 f.$$

All equalities are equalities of actual functions on S.

The strict set-colimit property of Lemma C.20 now gives a unique function

$$\rho : Z_k \longrightarrow \mathbb{R}, \qquad \rho\alpha_i = a_i.$$

Order the finite set Z_k first by its ρ-values and then by any chosen linear order within each fiber of ρ. If $t < t'$ in T, then $a_i(t) < a_i(t')$, so $\rho\alpha_i(t) < \rho\alpha_i(t')$. Hence this order makes every α_i strictly increasing. Possible equal values belonging to different copies cause no problem, since no two points of one copy have equal ρ-values. This proves the proposition. □

Remark C.22 (Proved scope and effective parameter selection)**.** The proposition asserts existence for all sufficiently large k; it does not assert orderability at every minimally admissible value. Its proof is a strict set-colimit argument and uses no conversion of a homotopy-coherent cocone into a strict one.

The choice can be made effectively. Place T at integer coordinates and choose the piecewise-linear h just described, with rational coefficients. Its fixed intervals and the endpoints of the other components are determined by finitely many affine equations. Iterate the finitely many rational points $g(x)$ and increase k until (S) holds, together with all additional numerical lower bounds. The convergence argument proves termination. The subsequent prescriptions for w and ρ are finite and rational. Thus arbitrary tie-breaking in the last step may also be replaced by a fixed deterministic ordering rule.

C.6 A choice of large repaired targets with persistent second homology

Lemma C.23 (Proved: convexity after a sufficiently long attachment)**.** *Let $f, g : S \rightrightarrows T$ be a parallel pair at a finite stage of the repaired AKR construction. Suppose that each of the induced graph embeddings $G_S \to G_T$ is injective on connected components and that the image of each component is geodesically convex. Here geodesically convex means that every shortest graph path between two of its vertices lies in that image. Let D be the largest diameter of a component of the finite graph G_T, taking the diameter of a singleton to be zero. Choose the repaired path bound $n \geq 2$ and a parameter $s = 4k + 1$, $k \geq n$, such that*

$$s > (2n - 2)(D + 1). \tag{LC}$$

Then every generating map $\alpha_i : T \to Z$ of the repaired target has geodesically convex image on each component of G_T. This lemma alone does not assert injectivity on components of $G_T \to G_Z$.

Proof. We first record the elementary linear-amalgam fact used below. If two graphs are glued along a graph whose maps into both pieces are injective on components and have geodesically convex component images, then each component of either piece is geodesically convex in the amalgam, and the two piece maps are injective on components. Indeed, a component of either piece contains at most one component of the gluing graph. Thus the component incidence graph of this gluing is a matching: each nontrivial resulting component is one component of

each piece glued along a single connected convex subgraph. Any excursion of a shortest path from one piece into the other has endpoints in that common subgraph. If the excursion were shortest between its endpoints, convexity in the other piece would place it in the common subgraph; if it were not shortest, replacing it would shorten the original path. Either possibility rules out a shortest excursion outside the original piece. The component and convexity claims follow. Induction applies this fact to a linear sequence of copies of G_T, glued along copies of G_S by any of the neighboring pairs $(g, f), (g, g), (f, g), (f, f)$. Every copy is componentwise convex and injective on components in that linear amalgam. At each induction step the image of the next gluing component is convex in the previous amalgam by transitivity of convexity, and is still injective on components.

Now fix a generating copy, two vertices in one of its source components, and a shortest path P between their images in G_Z. The path in the old component gives $|P| \leq D$. As in the cell-fiber proof of Proposition C.10, a target vertex is visible either in every copy, or in a proper cyclic interval of indices. The explicit repaired path formulas bound the size of every proper such interval by $2n - 2$; a private point has a singleton interval. Explicitly, on an actual path of $l \geq 2$ vertices the two middle visibility ranges have size l, and the two folded ranges have sizes $2(u + 1)$ for $0 \leq u \leq l - 2$ and $2(4k - u + 1)$ for $4k - l + 2 \leq u \leq 4k$. Each is at most $2l - 2 \leq 2n - 2$. There are at most $D + 1$ vertices of P. By (LC), the union of all proper visibility intervals of these vertices misses an index r. Vertices visible in every copy are deliberately omitted from this union.

Delete the one gluing overlap from copy $r - 1$ to copy r, with indices taken modulo s. The remaining diagram is a linear chain containing all s copies of G_T. Let L be its graph colimit. Its canonical map $q : L \to G_Z$ is the quotient which restores the omitted overlap. For every vertex of P, its cell-fiber graph remains connected after that overlap is deleted: a proper visibility interval avoids r, so it has no edge crossing this cut; a full circular fiber becomes a path. The same is true for every edge of P. Its visibility set is an interval or the entire index cycle. If either endpoint has proper visibility, the edge's interval lies in that endpoint's interval and avoids r. If both endpoints have full visibility, the routing invariant (c_{m+1}) factors the edge through every copy, since its two endpoints lie in that copy's image. Its fiber is circular and again becomes a path.

The preimages of a cell of G_Z in L are exactly the components of that cell's fiber after the indicated gluing edge is removed. Thus each vertex and edge used by P has a unique preimage in L. These preimages respect incidence, because the quotient is a graph map and the endpoint preimages are unique. They give a path $\widetilde{P}$ of the same length in L, whose endpoints are the original vertices of the selected copy of G_T.

The selected copy component is convex in L by the linear-amalgam fact. Its intrinsic distance between those endpoints is therefore their distance in L. This distance is at most $|\widetilde{P}|$. On the other hand, mapping an intrinsic shortest path to G_Z shows $|P|$ is at most that same intrinsic distance. The two lengths are equal, so $\widetilde{P}$ is shortest in L and convexity places it entirely in the selected copy. Mapping back by q places P in that copy's image. This proves the claim for every shortest path. □

Theorem C.24 (Proved: a selected repaired AKR category is not total)**.** *For every sequence (a_m) satisfying the numerical requirements of the repaired AKR Construction C.1, one can choose its finite targets so that all of the following hold at every stage:*

1. *the underlying point sets X, and separately the sets $H(X) = \pi_0 G_X$, admit linear orders for which every arrow induces an increasing injection on each of these two sets;*
2. *every arrow image component in G_X is geodesically convex;*
3. *every attachment induces an injection on the second integral homology of the binary category of elements.*

For the resulting countable union $A = \bigcup_m K_m$, the action

$$H = h_V/(bv_1 \sim bv_2)$$

satisfies

$$H_2(B\,\mathrm{El}_A\, H; \mathbb{Z}) \neq 0.$$

In particular there exists a quasi-filtered category which is not total. This is an existence theorem about selected sufficiently large targets; it does not assert the universal choice statement of Part C.1.

Proof. Initially G_V is a point and G_W is one edge and one isolated vertex. Give their underlying point sets and their component sets arbitrary linear orders. Every initial arrow is then increasing on points and on components, injective on components, and componentwise convex. The point order and component order are auxiliary independent orders; the quotient map $X \to H(X)$ is not required to preserve them.

Assume these assertions at a finite stage K_m, and consider one parallel pair $f, g : S \rightrightarrows T$. By the induction hypothesis both the point maps $f, g : S \rightrightarrows T$ and the component maps $H(f), H(g) : H(S) \rightrightarrows H(T)$ are increasing injections of finite linear orders. Neither partial transition relation has a nontrivial directed cycle: on the finite set of vertices of such a cycle the induced increasing bijection would be an increasing permutation, hence the identity. Both relations therefore consist of fixed points and finite directed paths. More generally every closed morphism-chain relation is a composition of restrictions of increasing maps and their increasing partial inverses. It is an increasing partial bijection on a finite order and has the same boundedness property. Thus the AKR property (4) holds here directly from these auxiliary orders.

Choose an odd integer $n \geq 3$ at least as large as the number of vertices in every nontrivial point path and every nontrivial component path. Apply Proposition C.21 to the ordered component pair and to the underlying ordered point pair. For each it permits every sufficiently large k and orders the strict set colimit of its repaired necklace so that all canonical copies are increasing injections. Choose k beyond both thresholds, and still larger if necessary so that (LC) holds and so that

$$s = 4k + 1, \qquad s/(3n) \in \mathbb{N}, \qquad \frac{s}{3n} \geq \frac{1}{1 - a_{m+1}/a_m}.$$

This is possible: since $3n$ is odd, the integers s divisible by $3n$ and congruent to 1 modulo 4 form an unbounded arithmetic progression. The ordinary finite repaired target with these parameters is an admissible target of C.1. Its point-set order is the order supplied by Proposition C.21 and Lemma C.20 for the point pair; all α_i are increasing as actual point maps.

We check that the component order just chosen is an order on the actual $H(Z)$. The vertex and edge fiber calculation in Proposition C.10 identifies G_Z with the strict graph colimit of the neighboring diagram of G_S and G_T. Graph components are computed by the coequalizer of the two endpoint maps from the edge set to the vertex set. Since colimits of sets commute with colimits, taking components preserves this graph colimit. Hence $H(Z)$ is the strict set colimit of the same necklace of $H(S), H(T)$. Lemma C.20, applied to their actual component path sizes, identifies this with the component target ordered above. Transfer that order along this canonical bijection. The maps $H(\alpha_i)$ are increasing injections.

Lemma C.23 supplies convexity of every component image of α_i. Every other incoming arrow is $\alpha_i b$ for an old arrow b, so increasing component injectivity follows by composition, actual point monotonicity also follows by composition, and componentwise convexity follows by transitivity. For a weak disjoint sum use the disjoint union graph and concatenate the two old orders, separately for points and for components. Its incoming maps satisfy both

assertions. There are only finitely many pairs and sums at a finite stage; make these choices for all of them, using fresh targets as the construction prescribes. No maps between distinct new targets are added. This proves the two geometric assertions at every stage.

A deterministic instance. To remove all choices from one counterexample, fix

$$a_0 = 2, \qquad a_m = \frac{3}{2} + \frac{1}{10 \cdot 2^{m-1}} \quad (m \geq 1).$$

These numbers strictly decrease and remain above $3/2$. The required reciprocals are 5 at $m = 0$ and $15 \cdot 2^m + 2$ at $m \geq 1$, hence integers. Initially label V and W by $0, 1$, label their points by integers, order the three points of W by $v_1 < v_2 < v_3$, and order its two graph components with the edge component first. Enumerate all subsequent objects, points, arrows, and ordered pairs lexicographically by their finite construction tags; arrows are stored as their tuples of point values.

At stage m, take the least odd $n \geq 3$ bounding every point-path and component-path size for every parallel pair of the finite stage. Let D be the largest diameter of any of its graph components. Test $k = n, n+1, \ldots$ in order. Retain only those satisfying the three displayed C.1 arithmetic conditions and $4k + 1 > (2n - 2)(D + 1)$. For each retained k, form the finite repaired target for every ordered parallel pair. On each target point set form the precedence graph with arcs

$$\alpha_i(x) \longrightarrow \alpha_i(y) \quad \text{for all } i \text{ and all } x < y \text{ in } T;$$

on its actual component set form the analogous graph with arcs $H(\alpha_i)(C) \to H(\alpha_i)(C')$ for $C < C'$ in $H(T)$. A loop is counted as a directed cycle. Choose the first retained k for which all these finitely many precedence graphs are acyclic. The two orderability applications prove that this search terminates. Assign each resulting point set and component set its lexicographically least topological ordering, always selecting the least available tag at a tie. This is a prescribed finite algorithm. It can use one common n, k for all pairs at the stage, so it also covers that more restrictive parameter convention.

Fixed-point, private-point, and path-coordinate tags specify the covered repaired target uniquely; path components are indexed by their least old point tag. Give every target a fresh tag consisting of its stage, construction type, source and target tags, and ordered pair of arrow tuples. For each ordered pair of old objects also add the freshly tagged disjoint-sum target, with left/right point tags and the concatenated point and component orders. Keep the old hom-sets, insert the prescribed generating maps, and close under composition as actual functions, identifying equal function tuples with the same source and target. These finite rules specify all objects and arrows. Composition is ordinary function composition, so it is associative and has the indicated identity functions. Each stage is finite, hence the union is a countable small category. The following homology argument applies to this fully specified instance as well as to the preceding existence construction.

For every parallel pair at such a stage, Lemma C.16 now applies: its old images are convex, its component maps are injective, and actual point monotonicity excludes a nontrivial point-transition cycle. Its fixed subgraph therefore meets each source component in an empty or connected subgraph. The exact kernel formula of Proposition C.10 shows that adjoining the pair's quasi-filter target kills no old second homology class. A disjoint-sum target also kills none: its cycle group is the direct sum of the cycle groups of its two summands, so the map a in Proposition C.5 is injective. Adding the finitely many new targets one at a time proves injectivity across every full stage.

Finally the first stage already has a nonzero class, independently of how large these first choices were. The two quasi-filter targets P_+, P_- for (v_1, v_2) and (v_2, v_1) each have one binary graph cycle; orient their generators κ_+, κ_- so that their augmentations are 1, as follows from

the signed neighboring count $2k-(2k-1)=1$. All other first-stage graphs are forests. For a pair involving v_3 and exactly one of v_1, v_2, every binary edge has its other endpoint private to that copy, so it cannot belong to a cycle. For a repeated point map the binary graph is a star, a disjoint union of edges, and isolated vertices. The identity quasi-filters and the initial weak sums also give forests. There are no arrows between distinct new first-stage targets, and the old cycle groups at V, W are zero. Consequently

$$\operatorname{colim}_{K_1} K = \mathbb{Z}\kappa_+ \oplus \mathbb{Z}\kappa_-, \qquad H_2(B\operatorname{El}_{K_1} H;\mathbb{Z}) = \ker(\mathbb{Z}^2 \xrightarrow{(1,1)} \mathbb{Z}) \cong \mathbb{Z}.$$

The generator $\kappa_+ - \kappa_-$ survives every subsequent stage. Every finite nerve chain of the union occurs in some stage, so its normalized chain complex is the filtered union of the stage chain complexes. Filtered unions of abelian groups are exact; therefore homology commutes with this union, and the persistent generator is nonzero in the displayed H_2 of A. The repaired construction is quasi-filtered, whereas totality would force this category of elements to have zero positive homology. Hence A has the claimed properties. □

C.7 Finite tests for totality: Lead 6

Lemma C.25 (Known, with the reduction proved: finite-poset representatives)**.** *Let C be a small category and let $n \geq 1$. Every free homotopy class of maps $S^n \to |NC|$ has a representative $|Nv| : |ND| \to |NC|$, after a homeomorphism $|ND| \cong S^n$, where D is the poset of nonempty faces of a finite triangulation of S^n and $v : D \to C$ is a functor.*

Proof. We recall the simplicial approximation argument, including the change of domain that is needed here. Choose a finite ordered simplicial complex K triangulating S^n. The natural weak equivalence $NC \longrightarrow \operatorname{Ex}^\infty NC$ has Kan target. Consequently a homotopy class $|K| \to |NC|$ is represented by a simplicial map $f : K \to \operatorname{Ex}^\infty NC$. Since K has finitely many nondegenerate simplices, f factors through $\operatorname{Ex}^r NC$ for some finite r. The subdivision–extension adjunction gives a simplicial map $g : \operatorname{Sd}^r K \to NC$. Under realization the adjunction compares $|g|$ with the original class by the last-vertex maps $\operatorname{Sd}^r K \to K$; their realizations are homotopic to the subdivision homeomorphisms. We may increase r to ensure $r \geq 1$. Now $\operatorname{Sd}^r K$ is the nerve of the poset D of nonempty faces of the finite complex $\operatorname{Sd}^{r-1} K$. The nerve is fully faithful on categories: the images of vertices and edges of g specify objects and arrows, and its two-simplices impose precisely the composition equalities. Thus $g = Nv$ for a unique functor $v : D \to C$. The facts about Ex^∞ and last vertices used in this argument are (Goerss and Jardine, 2009, Theorem I.11.4, Section III.4, and Theorem III.4.8); they concern arbitrary simplicial sets and do not require NC to be a simplicial complex. □

Theorem C.26 (Proved: Lead 6)**.** *Let A be a small nonempty connected category. For a functor $u : D \to A$, define the covariant A-set*

$$H_u(a) = \pi_0(u \downarrow a), \qquad H_u(b)([d,f]) = [d, bf]$$

for $b : a \to a'$. Here (d,f) has $f : u(d) \to a$, and an arrow $(d,f) \to (d',f')$ is $\delta : d \to d'$ with $f'u(\delta) = f$. Define a functor

$$j_u : D \longrightarrow \operatorname{El}_A(H_u), \qquad d \longmapsto (u(d), [d, 1_{u(d)}]),$$

whose value on an arrow is its image under u. The following conditions are equivalent:

1. *Every component of every $\operatorname{El}_A(H)$, for every covariant $H : A \to \mathsf{Set}$, is weakly contractible.*
2. *For every finite poset D and every $u : D \to A$, Bj_u is null-homotopic on every connected component of BD.*

3. *The same conclusion holds when D is the face poset of a finite triangulation of S^n, for every integer $n \geq 1$.*

In particular the tests in (3) suffice; no bound on the dimension of the spheres is being imposed.

Proof. The arrow $u(\delta) : u(d) \to u(d')$ defines an arrow in the displayed category of elements because $(d, u(\delta))$ and $(d', 1)$ are joined by δ in $u \downarrow u(d')$. This proves that j_u is well-defined.

The essential factorization is strict, not merely a homotopy factorization. Let $H : A \to \mathsf{Set}$ and let $v : D \to \mathrm{El}_A(H)$ be a functor. Write $v(d) = (u(d), h_d)$, where $u = \pi v$ and $\pi : \mathrm{El}_A(H) \to A$ is the projection. Define

$$\eta_v : H_u \longrightarrow H, \qquad (\eta_v)_a([d, f]) = H(f)(h_d).$$

If $f'u(\delta) = f$, then $H(f)(h_d) = H(f')H(u(\delta))(h_d) = H(f')(h_{d'})$, so this formula is constant on components of $u \downarrow a$. It is natural in a because $H(b)H(f)(h_d) = H(bf)(h_d)$. It therefore induces a functor over A,

$$E_v : \mathrm{El}_A(H_u) \longrightarrow \mathrm{El}_A(H), \qquad (a, \xi) \longmapsto (a, (\eta_v)_a(\xi)),$$

which is the identity on underlying arrows of A. Direct substitution shows $E_v j_u = v$ as functors.

Assume (1). Each component of the target of Bj_u is a weakly contractible CW complex and hence contractible by Whitehead's theorem. The continuous image of a component of BD lies in one such component, so the restricted map is null-homotopic. This proves (2), and (2) implies (3).

Assume (3), fix H and a component C of $\mathrm{El}_A(H)$, and consider any map $f : S^n \to BC$, $n \geq 1$. By Lemma C.25, after identifying the source with BD its free homotopy class is represented by Bv, for a functor $v : D \to C$. The preceding factorization gives $Bv = BE_v\, Bj_{\pi v}$. Condition (3) makes this composite null-homotopic. Thus every unbased sphere map into the connected space BC is null. This implies the vanishing of every based homotopy group: for $n = 1$ a free null-homotopy makes a loop conjugate to the identity and hence trivial; for $n > 1$ the change of basepoint acts by automorphisms and fixes the zero class. Therefore BC is weakly contractible. The case $H = \varnothing$ has no components and imposes no extra condition. Empty or disconnected D causes no difficulty in (2), which is expressly componentwise. □

C.8 The opposite interval-type category: Lead 8

Proposition C.27 (Proved: precise variance and a homological obstruction)**.** *Let* Ty *be the category in PRETOT Definition E.1: its objects are finite words τ in $+, -$, including the empty word, and a morphism $\tau \to \rho$ is an endpoint-preserving functor $I_\tau \to I_\rho$ whose function on vertices is nondecreasing. Here I_τ is the free category on the oriented line prescribed by τ. Then $\mathsf{Ty}^{\mathrm{op}}$ is isomorphic to the category of these words and nondecreasing sign-preserving maps of their letter sets.*

Put $p = (++)$, $q = (+-+)$, and use one-based letter indices. The arrows

$$e = (1, 3), \quad c = (1, 1) : p \rightrightarrows q$$

have no completion to a commutative square. For every nonzero unital commutative ring R they have no affine equalizer in $\mathrm{Aff}_R(\mathsf{Ty}^{\mathrm{op}})$. In fact, for the covariant set quotient

$$H = \mathsf{Ty}^{\mathrm{op}}(p, -)/(ge \sim gc)$$

we have $H_1(B\,\mathrm{El}_{\mathsf{Ty}^{\mathrm{op}}}(H); \mathbb{Z}) \cong \mathbb{Z}$. Thus $\mathsf{Ty}^{\mathrm{op}}$ is neither total nor exact. The phrase "over any ring" in Lead 8 must exclude the zero ring; with that qualification the lead is true.

Proof. For τ of length n and ρ of length m, write a type map on vertices as $\phi : \{0, \dots, n\} \to \{0, \dots, m\}$, with $\phi(0) = 0$, $\phi(n) = m$ and $\phi(i-1) \le \phi(i)$. The functor condition says exactly that every letter of ρ in the interval $(\phi(i-1), \phi(i)]$ has the sign of the ith letter of τ; a collapsed interval imposes no condition. Define

$$f_\phi : \{1, \dots, m\} \longrightarrow \{1, \dots, n\}, \qquad f_\phi(j) = \text{the unique } i \text{ with } \phi(i-1) < j \le \phi(i).$$

This is nondecreasing and sign-preserving. Conversely such a map f defines $\phi_f(i) = |\{j : f(j) \le i\}|$. These two formulas are inverse. They also cover the empty words: the map from a nonempty letter set to an empty one does not exist, while the map from an empty letter set is unique. If $\psi : I_\rho \to I_\sigma$ is another type map, the defining interval inequalities give $f_{\psi\phi} = f_\phi f_\psi$. Therefore these bijections form the asserted isomorphism of categories with the opposite variance.

Let $g : q \to \sigma$ be any arrow in the opposite category. Since adjacent letters of q have opposite signs, sign preservation and nondecreasingness imply $g(1) < g(2) < g(3)$. Hence ge is injective on the two letters of p, whereas hc is constant for every $h : q \to \sigma$. Thus $ge = hc$ is impossible, proving failure of condition I.

For each σ define an R-linear functional

$$\lambda_\sigma : R[\mathsf{Ty}^{\mathrm{op}}(p, \sigma)] \to R$$

by giving a basis arrow value 1 if it is injective and 0 if it is constant. These are the only possibilities for a map out of p. For every $g : q \to \sigma$ we have $\lambda_\sigma(ge - gc) = 1$. Consequently for every finite linear combination $t \in R[\mathsf{Ty}^{\mathrm{op}}(q, \sigma)]$,

$$\lambda_\sigma(t(e - c)) = \mathrm{aug}(t).$$

If $te = tc$, its augmentation is zero. It cannot also be 1 unless R is the zero ring. This is an exact integral dual certificate, valid for all targets and hence for all nonzero coefficient rings.

For the stronger homological claim, work over $\mathbb{Z}$ and put $P_x = \mathbb{Z}[\mathsf{Ty}^{\mathrm{op}}(x, -)]$, $d : P_q \to P_p$, $d(t) = t(e - c)$, $K = \ker d$, and $J = \operatorname{im} d$. The objectwise free-group functor preserves the defining set coequalizer, so $\operatorname{coker} d = \mathbb{Z}[H]$. Representables are projective and their colimits are canonically $\mathbb{Z}$ by augmentation. The functional identity shows that the map $\operatorname{colim} K \to \operatorname{colim} P_q = \mathbb{Z}$ is zero, even when its input is a finite sum of elements represented at different objects. The exact sequence $K \to P_q \to J \to 0$ therefore gives $\operatorname{colim} J \cong \mathbb{Z}$. The map $\operatorname{colim} J \to \operatorname{colim} P_p$ is zero because the augmentation of every $t(e - c)$ is zero. The derived-colimit exact sequence of $0 \to J \to P_p \to \mathbb{Z}[H] \to 0$ now gives

$$L_1 \operatorname{colim} \mathbb{Z}[H] \cong \operatorname{colim} J \cong \mathbb{Z}.$$

The bar complex for the permutation diagram $\mathbb{Z}[H]$ is exactly the integral nerve chain complex of its category of elements, proving the claim. This explicitly witnesses failure of both totality and exactness and does not infer an infinite conclusion from finite tests.

The program `code/finite_ty_verify.py` independently checks the two descriptions on all 961 ordered pairs of words of length at most four, the contravariant composition bijection on 2731 composable pairs of type maps for words of length at most three, and the integral dual certificate on all 7937 arrows from q to the 1023 words of length at most nine. These computations audit the formulas; the preceding proof applies without a length bound. □

C.9 Constraints on a minimal separator: Lead 9

Proposition C.28 (Proved: Lead 9)**.** *Suppose A is a small total category that is not exact, and let $a_0, \dots, a_k : p \to q$ be a family without an integral affine equalizer. Assume k is least among all such families in A. Then:*

1. $A(q,p) = \varnothing$.
2. *The finite cardinalities* $|A(p,x)|$ *have no uniform finite bound; in symbols,* $\sup_x |A(p,x)| = \infty$ *in the sense of unboundedness in the natural numbers, allowing infinite values.*
3. *The full cosieve* $V_p = \{x \in A : A(p,x) \neq \varnothing\}$ *is again a separator, with the same exhibited family. Its object* p *is weakly initial, and the minimal integer* k *remains the same after restriction.*
4. *For every arrow* $g : q \to x$*, the arrows* $ga_0, \dots, ga_k : p \to x$ *are pairwise distinct.*

Proof. Totality gives conditions I and IV. We use precisely the two exclusions proved in the opened source (Isbell and Mitchell, 1976, Corollary 4.3 and Proposition 4.5). Corollary 4.3 first simultaneously equalizes a finite collection of endomorphisms with the identity, by applying the fixed-point theorem to endomorphisms of a representable act. If $b : q \to p$ existed, apply it to $ba_0, \dots, ba_k, 1_p$. It gives an ordinary arrow $u : p \to r$ with $uba_i = u$ for every i. The ordinary arrow $ub : q \to r$ would already be an affine equalizer, proving (1).

For (2), the representable A-set $A(p,-)$ is connected because its category of elements has the initial object $(p, 1_p)$. If its values had a uniform finite bound, Proposition 4.5 would imply condition II for this act. That proposition forms the congruence identifying two elements whenever an ordinary arrow equalizes them; condition I makes this a congruence. All arrows act injectively on its quotient. At an object where the finite cardinality is maximal, the maps to subsequent objects are bijections, and the necklace form of IV forces the quotient to have one element there. The conclusion is exactly ordinary equalizability of every pair in the original representable. Apply it successively to the finitely many images of $a_0, \dots, a_k$; composing the resulting ordinary arrows gives an ordinary simultaneous equalizer, a contradiction. This verifies that the cited result applies to the one representable involved here, rather than requiring a bound on every hom-set of A.

For (3), V_p is nonempty and is closed under targets. It is connected: condition I applied to a span $p \to x$, $p \to y$ gives a common target inside V_p. A covariant set-valued functor on V_p extends to A by the empty set off V_p; arrows entering V_p have the unique map from an empty set, and no arrow leaves it. Its category of elements is literally unchanged. Hence totality of A proves totality of V_p. Any affine equalizer of the displayed family in V_p would also be one in A, so V_p is not exact. Conversely, if a family of arrows in V_p has an affine equalizer in A, every target occurring in its finite support is in V_p, because it receives an arrow from its source in the cosieve. Thus it is already an affine equalizer in V_p. All smaller families therefore remain equalizable. Every object of V_p receives a map from p, which is exactly weak initiality. This is the full proof of the cosieve step used from TOTEX D.23.

For (4), suppose $ga_i = ga_j$ for some $i \neq j$. The displayed family at x then has at most k distinct arrows. By the global minimality of k there is an affine $t : x \to y$ equalizing all its distinct members. The composite $tg \in \mathbb{Z}[A(q,y)]$ has augmentation one, because g is an ordinary arrow, and satisfies $(tg)a_i = (tg)a_0$ for every i. This contradicts the choice of the original family. This argument uses minimality among all objects and families of A, as stipulated in the lead; minimality merely within a fixed target would not suffice. □

D Class diagram, manuscript readings, and sources

D.1 What the class diagram currently decides

Proposition D.1 (Proved: the class diagram with strict right inclusion)**.** *Among nonempty connected small categories,*

$$\{\textit{filtered}\} \subsetneq \{\textit{exact}\} \subseteq \{\textit{total}\} \subsetneq \{\textit{quasi-filtered}\}.$$

The ordinal-injection category witnesses the first strict inclusion. The selected repaired AKR category of Theorem C.24 witnesses the last strict inclusion. Whether the middle inclusion is an equality remains undecided here.

Proof. Filtered colimits of abelian groups are exact. The complex computing $T_i(\mathbb{Z}[H])$ is the integral nerve-chain complex of $\mathrm{El}_A H$, so exactness implies acyclicity of every component. We check the homotopical step in supplied TOTASPH C.3–C.6. For a component $E \subseteq \mathrm{El}_A H$ and $S : E \to \mathrm{Set}$, the functor

$$a \longmapsto \coprod_{(a,h)\in E} S(a,h)$$

has category of elements isomorphic to $\mathrm{El}_E S$. Hence acyclicity also holds for actions on E. For any cyclic subgroup $C \leq \pi_1(BE, e)$, the action $x \mapsto \mathrm{Hom}_{\Pi E}(e, x)/C$ gives the connected covering with fundamental group C, as is checked by edge-path lifting. Its first homology is C, forcing $C = 0$. Thus BE is simply connected, and Hurewicz turns its acyclicity into weak contractibility. This does not replace permutation modules by arbitrary diagrams.

For totality implying I, the checked TOTASPH C.11 argument first gives common successors. If two principal cosieves were disjoint, identify the values of the constant two-element action on their union. Its element category is the double of A identified on those cosieves. Its reduced homology sequence has $H_1 = \mathbb{Z}$, since BA is contractible and the two cosieves are nonempty and connected. This contradicts totality. Applying common successors to the inherited total category $p/A = \mathrm{El}_A A(p, -)$ completes every span out of p to a commuting square. The case $k = 1$ of Lemma B.2, followed by common successors, gives IV. Conditions I and IV give quasi-filteredness.

We checked the interpolation proof for Δ_{inj} in supplied TOTASPH C.8–C.10. Its identity $P_{t\delta^i} = y_i D_i P_t$ encodes gaps of injections as monomials. The integral coordinates $x_i = z_i - z_{i+1}$ reduce extension of compatible homogeneous restrictions to the alternating sum over nonempty coordinate intersections. The prescriptions $D_i C_{n,k} = C_{n-1,2k} \prod_{\ell \neq i} y_\ell^k$ agree on intersections and have value one at the all-ones point. Thus they give an augmentation-one combination equalizing the cofaces; the simplicial identities then equalize all parallel injections. Theorem 1 and Lemma 1 of Isbell and Mitchell (1973), or Theorem 3.1 and Proposition 3.3 of Isbell and Mitchell (1976), give exactness. The two point inclusions $[0] \rightrightarrows [1]$ cannot be post-equalized by an injection. The 1973 Theorem 3 announces the example; the body of Isbell (1974) was not obtained and is not claimed as a proof checked in this run.

In repaired AKR, each parallel pair receives a quasi-filter and each object pair a common successor, by the opened Constructions B.9 and C.1 of Adámek et al. (1981). These give I and IV at the union: every finite diagram occurs at a finite stage. Proposition C.4 proves nonexactness for every admissible repaired construction. Theorem C.24 gives a deterministic choice of sufficiently large targets for which the binary category of elements has nonzero second homology. That category is quasi-filtered and is not total, proving the final strict inclusion without assuming the universal choice statement of C.1. □

Undecided statement D.2 (Open in this incomplete record: Part D.1)**.** We have not decided whether exactness equals totality. The other comparison is now decided: totality is strictly stronger than quasi-filteredness, by Theorem C.24. The selected AKR category is not total and therefore is not a separator for the remaining middle inclusion.

D.2 The corrected manuscript statements

Proposition D.3 (Refuted conjecture; known corrected implications)**.** *The implication* totalement W_∞-asphérique $\Longrightarrow$ filtrante *at II manuscript page 36ter is false. For nonempty connected small A,*

$$A \text{ filtered} \iff A \text{ total and sifted.}$$

For every fixed integer $n \geq 1$,

$$A \text{ filtered} \iff A \text{ total and } \operatorname*{hocolim}_A K(M(a), n) \text{ is } n\text{-truncated for every covariant } M : A \to \mathrm{Ab}.$$

The implication from siftedness alone to totality in Proposition 2.19 at II manuscript page 36 is false.

Proof. Proposition D.1 refutes the conjecture. We read the wording in the supplied transcription:

> "Cela me fait soupçonner que la conjecture"
>
> $$X \text{ totalement } W_\infty\text{-asphérique} \Longrightarrow X \text{ filtrante}$$
>
> "pourrait être valable."

We do not substitute an unproved equivalence with exactness.

For the first corrected equivalence, we checked supplied PRETOT F.3 and TOTEX D.4 and the finite cocone criterion in Gabriel and Ulmer (1971, Sections 15.2(c), 15.9). Siftedness means $C_{p,q} = \mathrm{El}_A(A(p,-) \times A(q,-))$ is nonempty and connected. Inherited totality gives common successors in $C_{p,q}$. For $u, v : p \to q$, a successor of $(q; u, \mathrm{id}_q)$ and $(q; v, \mathrm{id}_q)$ gives $f, g : q \to r$ with $fu = gv$ and $f = g$. Thus f post-equalizes u, v. Siftedness also gives common successors in A, so A is filtered. The converse is the filtered-colimit criterion.

We checked the spectral-sequence steps of supplied TOTEX D.5, D.7 and D.9; the earlier abelian one-type argument is PRETOT F.6. For $n \geq 2$, BA is contractible and $X_M = \operatorname{hocolim}_A K(M, n)$ is $(n-1)$-connected. Its degree-n homology is $\operatorname{colim} M$, so the assumed truncation makes the canonical map $X_M \to K(\operatorname{colim} M, n)$ a weak equivalence. The reduced homology bar spectral sequence has first nonzero row M in degree n and zero row in degree $n+1$, since $H_{n+1}K(M,n) = 0$. Hence $T_1(M) = H_{n+1}(X_M) = 0$, first proving exactness. Collapse in all degrees, applied to $M \oplus N$, identifies the natural Künneth cross summand in degree $2n$:

$$\operatorname*{colim}_A (M \otimes N) \cong (\operatorname*{colim}_A M) \otimes (\operatorname*{colim}_A N).$$

There is no Tor term in that degree, since two positive indices are at least n. On P_p, P_q, this is the free abelian map from the component set of $C_{p,q}$ to a singleton. It is an isomorphism only if that set is a singleton. Thus A is sifted and hence filtered.

For $n = 1$, common successors make the group colimit of an abelian diagram abelian: any two representatives commute after transport to a common object. Since BA is contractible, the fundamental-group presentation of $\operatorname{hocolim}_A BM$ gives $G = \operatorname{colim}_A M$. The one-type assumption identifies it with BG. In degree two the homology edge image is that of $\operatorname{colim}_A \Lambda^2 M \to \Lambda^2 G$, with remaining filtration quotient $T_1(M)$. The edge map is surjective by transporting the two factors of an exterior product to a common successor, so $T_1(M) = 0$. After exactness is obtained, the cross summand of $\Lambda^2(M \oplus N)$ gives the same tensor comparison and siftedness. Conversely, filtered homotopy colimits preserve these truncations because spheres, nullhomotopies and horns involve finitely many simplices.

For Proposition 2.19 take $A = \Delta^{\mathrm{op}}$ and $H = \Delta[1]/\partial\Delta[1]$. In our covariant convention, $\mathrm{El}_A H$ is opposite to the usual simplex category, so its classifying space is weakly equivalent to $|H| = S^1$. Thus A is not total. It is sifted: the cospan category in Δ^{op} on $[m], [n]$ is opposite to the simplex category of $\Delta[m] \times \Delta[n]$, which is nonempty and connected. This checks the variance of the counterexample. □

Proposition D.4 (Known, with the comparison maps audited: positive truncations)**.** *For every small category A, including empty and disconnected categories, let $\ell : A \to \Pi A$ be groupoid completion. For every fixed finite $n \geq 1$, homotopy colimits over A preserve n-types if and only if*

$$C_a = a \downarrow \ell = \mathrm{El}_A(\mathrm{Hom}_{\Pi A}(a, \ell(-)))$$

is filtered for every a. This is supplied TOTEX D.10.

Audit of the relied-on reduction. An object of C_a is $(x, \gamma : a \to \ell(x))$; an arrow to (y, δ) is $f : x \to y$ with $\ell(f)\gamma = \delta$. All transition maps are bijective, and edge-path lifting identifies $BC_a \to BA$ as the connected universal cover of its component. For $Y : \mathrm{El}_A S \to \mathrm{Spaces}$, the functor $b \mapsto \coprod_{s \in S(b)} Y(b, s)$ has the same simplicial replacement as $\mathrm{hocolim}_{\mathrm{El}_A S} Y$. Preservation of n-types therefore descends to C_a.

The simply connected case in TOTEX D.9 uses the reduced homology bar complex. The constant point diagram first makes BC_a an n-type. For $n \geq 2$, the section of $\mathrm{hocolim}\, K(M, n) \to BC_a$ and the discreteness of the based self-equivalence space of $K(L, n)$ trivialize its Eilenberg–MacLane fibres over this simply connected base. Relative homology in degree $n + 1$ is zero and equals $T_1(M)$. For $n = 1$, the double-cosieve action excludes absence of common successors by a surviving $E^2_{1,1} = \mathbb{Z}$ term; the abelian one-type argument then applies. Thus C_a is exact, total and, by the fixed-n test, filtered. These are the specific steps checked in the supplied proof; we do not attribute the classification to a differently stated theorem on filtered colimits.

Conversely, filteredness makes each BC_a contractible. The pullback of $BC_a \to BA$ along $\mathrm{hocolim}_A X \to BA$ is modeled by $\mathrm{hocolim}_{C_a} X\pi$: on the bar construction, a lift of a chain is determined by a path class from a to its first vertex. Filtered homotopy colimits preserve n-types; covering maps induce isomorphisms on homotopy groups in degrees at least two. This proves the converse componentwise. For empty A both assertions hold vacuously. □

Remark D.5 (Proved: typing corrections and scope)**.** At II manuscript page 36, for $f : X \to Y$, finality must use $y\backslash X$ rather than the printed $y\backslash Y$. The latter always has initial object id_y and would make the condition vacuous. At XII manuscript page 88, $H' \in (U')^\wedge$, so the correctly typed isomorphism is

$$U'/t' \simeq U'/H',$$

not $U'/t' \simeq U/H'$. A presheaf on U is a covariant set-valued functor on U^{op}; its usual element category is opposite to the corresponding covariant element category. Opposition preserves the classifying-space homotopy type.

Definition 1 at XII manuscript pages 88–89 quantifies over all left Kan extensions and equivalently over ordinary final images. It is the notion used here, distinct from the test-category diagonal condition in *Pursuing Stacks*. We inspected the supplied transcription, not an independent copy of the authoritative edition. These are mathematical typing corrections, not a new editorial collation.

D.3 Literature and provenance

The access table in `TOTEX2_report.md` records every requested source, exact inspected theorem numbers, failed retrievals and search terms. We obtained Isbell (1968) and Gabriel–Ulmer (1971), which had not been opened in the supplied record. We did not obtain the bodies of Isbell (1974), Oberst (1968) or Adámek–Reiterman (1975). No claim about their unopened contents is used as a proof. Each bibliography entry records body-access status.

The coefficient-ring criterion is the remark on page 296 of Isbell and Mitchell (1976); the weak-terminal and bounded-hom-set exclusions are its Corollaries 4.3–4.4, Proposition 4.5, and Corollary 4.6. The equivalence printed in Adámek and Reiterman (1976, II.3) is false. Adámek

et al. (1981, Remark A.4) explicitly corrects the earlier 1975 remark; their counterexample also refutes the assertion repeated in the 1976 paper, which A.4 does not explicitly cite. The fixed-point/exactness question in Husainov (2002, Problem 2.5) has a weaker hypothesis than totality and does not decide our main question.

For monoid acts, preservation of monomorphisms and of finite limits are distinct meanings of flatness; Hemelaer and Rogers (2021, Definition 1.6, Proposition 1.8, Theorems 2.32 and 2.49) separates them. We also inspected Laan (nd, Proposition 2.4 and the paragraph after Lemma 2.6). Neither notion may be silently identified with flatness of the constant right $\mathbb{Z}A$-module.

We make no priority claim. Searches included exact colimits, fixed-point properties, homotopy-flat actions, totally aspherical categories, Ramsey classes, and exact titles and DOIs of the requested sources. No opened source in these searches supplied the missing totality-to-exactness implication or a counterexample. This reports the search performed, not a claim that the question is open in the mathematical literature.

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